



DCL-07

**DELTA, SIGMA, ADAPTIVE DELTA
MODULATION & DEMODULATION KIT**

EXPERIMENTAL MANUAL

.... A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. DIGITAL COMMUNICATION is one of the important subject need to teach while learning electronics.

It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

-Light years ahead – refers to leadership.

As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

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We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete comprehensive warranty.

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SAFETY RULES

Carefully follow the instructions contained in this manual as they provide you with important points on safety during the installation, use and maintenance. Keep this manual always with you for easy reference.

Arrange all accessories in order after unpacking, so that its integrity is checked with respect to its checklist. Also ensure that no visible damage as such appear on any accessories.

Before connecting the power supply to the kit, be sure that the jumpers and Connecting chords are connected correctly as per experiment.

In DCL-07 kit, signals are to be monitored with an oscilloscope as explained in the manual. For observation of signals on oscilloscope either use X10 (Attenuation Probe) or 180Ω resistance in series with normal oscilloscope probe.

This kit must be employed only for the use for which it has been conceived, i.e. as educational equipment, and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of any fault or malfunctioning in the trainer kit, turn off the power supply and do not tamper the kit. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the warranty period if the service is needed, purchaser should get in touch with the service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions. is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and upgradation of products, be sure to contact our service center or sales outlet.

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TECHNICAL SPECIFICATIONS

Signals

Internal frequency: 250Hz, 500Hz, 1KHz, and 2KHz.

Amplitude: 0 to 4V (p-p)

External frequency: Upto 2KHz. 0 to 4V DC Shifted.

Internal DC signal Amplitude: 0 to 5V

Sampling

Internal sampling clock: 8 KHz, 16 KHz, 32 KHz, 64 KHz and 128 KHz

Duty cycle: 50%

Amplitude: 5V.

Modulation Techniques

Delta Modulation

Adaptive Delta Modulation

Sigma Delta Modulation

CVSD Modulation

Compressor

Expander

Voice Modulation

Voice link with the help of microphone, speaker and using Noise and Audio Pre-Amplifier kit & Audio Amplifier kit (not provided along with the trainer) can be studied using different modulation techniques like Delta Modulation, Sigma Delta Modulation, Adaptive Delta Modulation and CVSD.

Low Pass Butterworth Filter

2nd Order and 4th order low pass butter-worth filter with cut-off frequency of 3.4KHz.

Interconnections

This is done with the help of supplied connecting chords and 2mm sockets.

Test Points

Various test points onboard avails the user to observe various signals on oscilloscope.

Switch Faults

Switch fault section provides 8 switches. These switches can be used to simulate fault conditions in various parts of the circuit.

Power Supply

-12V, +12V, +5V, GND

Accessories

DCL -07			Quantity
1	Mini Links PCS2 Patch cord	Male To Male	10
2	DCL-07 EXPT Manual		1
3	Power Supply		1
4	Power cord		1

FEATURES

- **Function Generator**

This section basically provides four Amplitude variable (0-4Vpp) synchronized sine waves each of 250Hz, 500Hz, 1KHz and 2KHz. The 2.048 MHz crystal Oscillator generates a 2.048 MHz clock. It is divided by 2 decade counters and 2 ripple counters to get the 32 KHz, 16 KHz, 8 kHz and 4 KHz frequencies. This signal is fed to, serial to parallel shift register, which generates the stair case sine wave, by serial shift operations. The serial to parallel shift register has a resistive ladder network at the output. For 16 shifts of the register, one stair case sine wave is produced. So if a 16 KHz clock is fed to the shift register, 1 kHz sine wave is generated. Thus by using the 32 KHz, 16 KHz, 8 kHz and 4KHz, sine wave of 2 KHz, 1 KHz, 500 Hz and 250 Hz are generated respectively. The active filter at the output suppresses the ripple and also takes care of the impedance matching. Filter output gives smooth sine waves.

A DC signal with Variable Amplitude (0-5Vpp) is also generated.

- **Sampling Clock Generator**

This section provides five different sampling frequencies of 8 KHz, 16 KHz, 32 KHz, 64 KHz and 128 KHz. All these frequencies are generated from main clock of 2.048MHz crystal. The LED shows the clock that has been selected for sampling during modulation.

- **Delta Modulator And Demodulator**

This is a process of converting Analog signal into one bit code. In Delta modulation only one bit is sent per sample. This bit indicates whether the signal is larger or smaller than the previous samples. The advantage of Delta modulation and Demodulation is, the circuits are much simpler than those used in traditional PCM. Some limitation of Delta Modulation is over come in Adaptive delta modulation systems. In encoding process, the logic levels of the transmitted pulses indicate whether the decoded output should rise or fall at each pulse. Delta Modulation is true digital encoding process compared to PAM, PWM and PPM. Delta Modulation system cannot react to rapid amplitude changes between samples. It is primarily used in voice communication.

- **Adaptive Delta Modulator**

As Delta Modulation system is unable to chase the rapidly changing information of the analog signal which gives rise to distortion & hence poor quality reception. This is known as slope overloading phenomenon. The problem can be overcome by increasing the integrator gain (i.e. step size). In Adaptive Delta Modulation the step size is not kept fixed, when slope overload occurs the step size becomes progressively larger. Adaptive Delta is similar to Delta Modulation except for the gain control circuit. Thus Adaptive Delta Modulation system is able to reduce slope overload error at an expense of small increase in quantisation error. The Adaptive Delta Modulator can operate at the rate of 32 Kb/s with performance comparable to that obtained using PCM at 64 Kb/S.

- **Sigma Delta Modulator**

The quantize (feedback) input in Delta Modulation is viewed as an approximation to the derivative of the incoming signal. This behavior leads to a drawback of delta modulation in that transmission disturbances such as noise result in an accumulative error in demodulated signal. This drawback is overcome by integrating the incoming signal prior to delta modulation. This way modulator now responds to the actual voltage levels rather than the change in voltage level. Integrator at input of Delta modulator is called Sigma

Delta modulator. Through Sigma delta modulator we can transmit DC signal and Receiver design becomes simple. We require only low pass filter at the receiver side.

- **Continuously Variable Slope Delta Modulator (CVSD)**

The amplifier has a variable gain i.e. its gain is function of the voltage applied at its gain control terminal. The gain increases with increase in positive gain control voltage. Many Adaptive Delta Modulators do not change step size on a pulse to pulse basis, rather changes are made much more slowly, such slow control are referred to as syllabic. The usual implementation involves a Continuously Variable gain circuit controlled by an analog voltage from step size controller. This is often called **Continuously Variable Slope Delta (CVSD)**. There is a variety of CVSD IC encoder and decoder in today's semiconductor market.

- **Companding**

A process called COMPANDING; a word formed by process of compressing and expanding, can materially improve the dynamic range. Companding is the process of compressing the signal amplitudes before transmission and expanding the compressed signal amplitudes after reception.

The dynamic range of Delta Modulation system is limited. The maximum input level is limited by slope overload and minimum input level is limited by quantization noise. The limited dynamic range of delta modulation can be increased by use of Companding.

- **Voice Modulation**

One has high pitched voice another low pitch. One enunciates clearly another slur words. To transmit speech we need not transmit the precise waveform generated by the speaker. We can transmit information from which a waveform can be reconstructed at the receiver, which is only similar to, rather than identical to the waveform generated by the speaker. There is an audio preamplifier before modulation process and audio amplifier after demodulation process. Voice can be modulated by DM, SDM, ADM, CVSD and companding process using Noise and Audio Pre-Amplifier kit & Audio Amplifier kit.

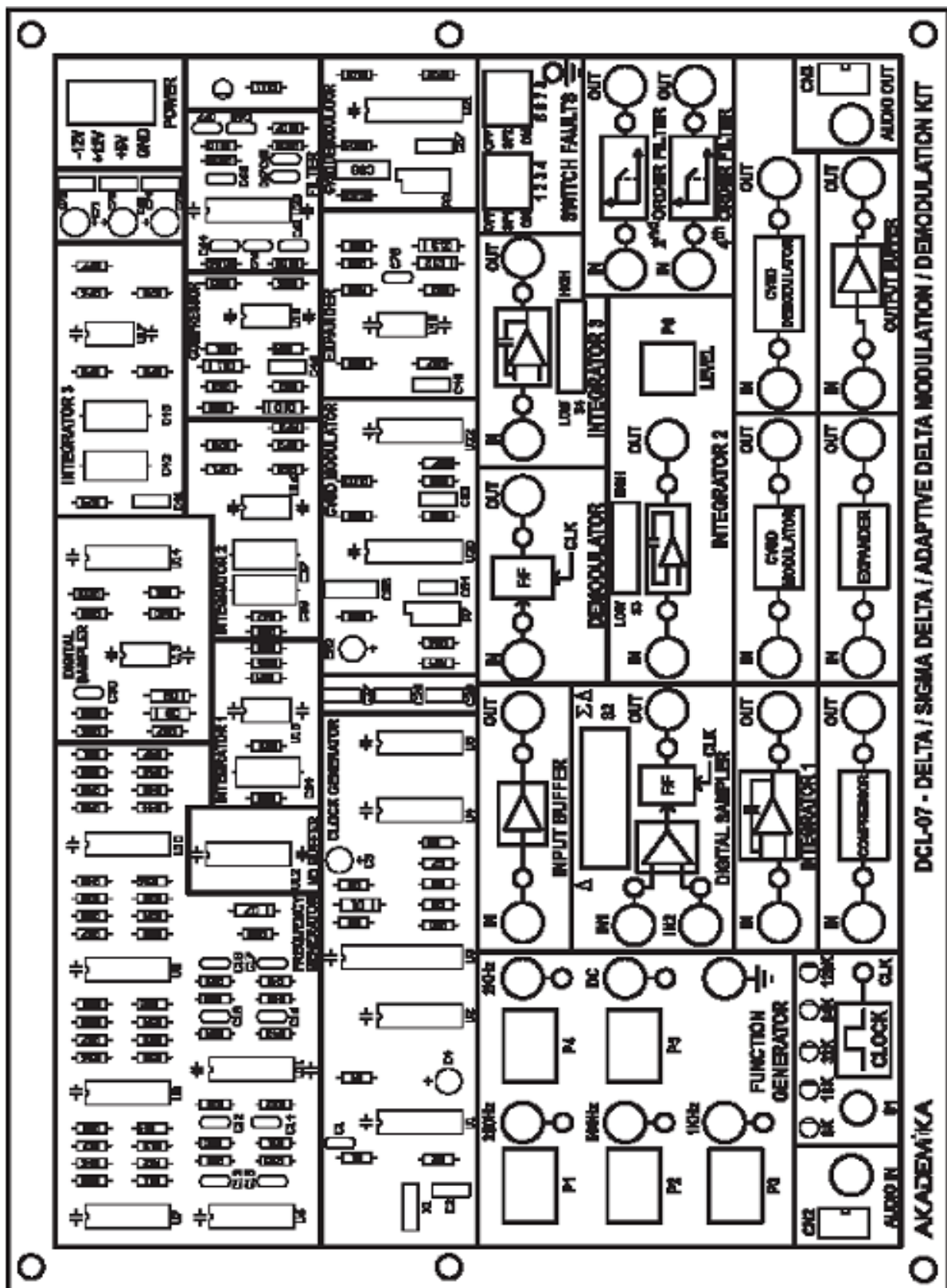
- **Filter Section**

In this section 2nd and 4th order low pass Butter worth Filter is provided which filters out the sampling frequency components from the Demodulated output of each modulation technique and recovers the original signals.

- **Switch Faults**

Switch fault section provides 8 switches. These switches can be used to simulate fault conditions in various parts of the circuit.

FUNCTIONAL BLOCK



INTRODUCTION

“ DCL-07: DELTA / SIGMA / ADAPTIVE DELTA MODULATION AND DEMODULATION KIT ” is a compact single board unit in wooden casing and it provides hands on experimentation in the area of Delta Modulation - Demodulation, Adaptive Delta Modulation - Demodulation, Sigma Delta Modulation – Demodulation, CVSD (Continuously Variable Slope Delta Modulator), Compressor and Expander and also Voice Modulation and Demodulation techniques.

This unit contains onboard function generator, which provides Synchronized amplitude variable sine waves of 250Hz, 500Hz, 1 KHz and 2KHz and DC signal with variable amplitude. The unit has provision to accept external sine wave. Selectable clock frequency for modulation makes this kit ideal for experimentation. All onboard signals can be easily interconnected with the help of connecting chords, this signals are phase locked to enable the learner to observe stationary signals. The trainer has complete circuit mimic inscribed on PCB.

In digital Modulation system, the analog signal is sampled and digitally coded. This code represents sampled amplitude of analog signal. The digital signal is sent to the receiver through any channel in serial form. At the receiver the digital signal is decoded and filtered to get reconstructed analog signal. A sufficient number of samples are required to allow the analog signal to be recovered accurately. Delta Modulation is one of the digital Modulation technique developed after Pulse Code Modulation, in which the difference signal $d(t)$ is encoded into just a single bit. If the signal amplitude increases then the Modulator output is a logic level 1. If the signal amplitude decreases the modulator output is a logic level 0. Thus the output of modulator is a series of ones and zeroes to indicate rise and fall of waveform with respect to the previous value

Circuit Description

Clock Generator

This section provides five different sampling frequencies. All these frequencies are generated from main clock of 2.048MHz crystal. The 2.048MHz crystal Oscillator generates a 2.048MHz clock. It is divided by 2 ripple counters U2(74LS393) to get five different sampling frequencies of 8KHz, 16KHz, 32KHz, 64KHz and 128KHz. The different clocks for sampling in modulation techniques can be selected with the help of U3 (GAL16V8-25LC), driving circuit for the U3 is setup by U5 (74LS393) and U4 (74LS74 – D flip flop) through switch S1. LED indications are provided for each selected sampling frequency using U3 (GAL16V8-25LC).

- **Sine Wave Generation**

This section basically provides four Amplitude variable (0-4Vpp) synchronized sine waves each of 250Hz, 500Hz, 1KHz and 2KHz. The 32KHz, 16KHz, 8 kHz and 4KHz frequencies are fed to serial to parallel shift register (74LS164), which generates the store case sine wave, by serial shift operation. The serial to parallel shift register has a resistive ladder network at the output. For 16 shifts of the register, one store case sine wave is produced. So if a 16KHz clock is fed to the shift register, 1 KHz store case sine wave is generated. Thus by using the 32KHz, 16KHz, 8 KHz and 4KHz, store case sine wave of 2KHz, 1KHz, 500Hz and 250Hz are generated respectively by U10, U9, U8, U7. The active filter at the output (LM324) suppresses the ripple and also takes care of the impedance matching. After filtering we will get smooth sine waves. Amplitude of the signal is varied using respective potentiometers.

- **DC Generation**

In this section Zener Diode D7 (5V1) will regulate VCC to 5 volts. Amplitude is varied by using potentiometer P5.

- **Input Buffer**

Refer to the circuit diagram as shown later in this manual, sheet 2 of 6 of **DCL-07**. This is a non-inverting unity gain amplifier built around U12 (LM324).

- **Digital Sampler**

Digital sampler consists of two stages. First stage is of differential amplifier U13(LM311) and second stage is D flip-flop U14(4013). Differential amplifier gives output by comparing voltage levels at input terminals. Output is equal to signal IN2 subtracted from signal IN1. D flip-flop sends this output at every rising edge of the clock applied to it.

- **Integrator-1**

Integrator is built around U15 (IC LF353). Integrator output is integration of the input signal applied.

- **Integrator-2**

Integrator is built around U16 (IC LF353). Integrator output is integration of the input signal applied. In this integrator switch S3 is provided to select low or high time constant for integration operation. Pot P6 is provided to control the gain of integrator.

- **Delta Or Sigma Delta Selection**

Switch S2 is provided for selection. In case of Delta Modulation signal is bypassed through Capacitor, DC signal is blocked. In case of Sigma Delta Modulation DC signal is also send to integrator inputs.

- **Integrator-3**

Integrator is built around U17 (IC LF353). Integrator output is integration of the input signal applied. In this integrator switch S4 is provided to select low or high time constant for integration operation.

- **Demodulator**

Demodulator is simple D flip-flop using U14 (4013). Output is available at rising edge of clock applied to it.

- **Compressor**

Compressor is Log amplifier built around U18 (LF353). For low amplitude signal output is linear and for high amplitude signal output is logarithmic.

- **Expander**

Expander is Antilog amplifier built around U19 (LF353). For low amplitude signal output is linear and for high amplitude signal output is logarithmic.

- **CVSD Modulator**

The CVSD is built around U20 (IC MC3418). The CVSD is the simple alternative to more complex conventional conversion techniques, in systems requiring digital communication of analog signals. The CVSD A/D is well suited for the requirements of digital communications. A Delta Modulator consists of a comparator in the forward path and an integrator in the feedback path of a simple control loop. The input to the comparator is the

simple analog signal and the integrator output. The comparator output is the difference between the input voltage and the integrator output. The sign bit is the digital output and also controls the direction of ramp in the integrator. The output of comparator is fed to the sampler. Then the sampler output is fed to the slope polarity switch and level detect algorithm. The level detect algorithm again fed to the slope magnitude control followed by slope polarity switch. The output slope polarity switch is fed to the integrator in the control loop. With no input at the transmitter a continuous 1 and 0 alternations are transmitted. The outstanding characteristic is its ability to transmit the intelligible voice out at relatively low data rate. Companded PCM for telephone quality transmission requires about 64Kbits/sec data rate per channel. CVSD produces equal quality at 32Kbit/sec.

- **CVSD Demodulator**

The CVSD Demodulator is built around U21 (IC MC3418). In CVSD Decoder CVSD mod output fed to the input of comparator. The comparator output is fed to the internal shift register. Then the output of internal shift register is fed to the digital logic followed by slope polarity switch and integrator. The output of integrator is fed to the low pass filters for the reconstruction of original signal.

- **Output Buffer**

This is a non-inverting unity gain amplifier built around U12 (LM324).

- **2ND Order Butterworth Filter**

Refer to the circuit diagram as shown later in this manual, sheet 6 of 6 of **DCL-07**. 2nd order Butter worth Low Pass Filter is built around U23 (IC LM324), which filters out the sampling frequency components from the Demodulation output of each modulation techniques and recovers the original signals.

- **4TH Order Butterworth Filter**

Refer to the circuit diagram as shown later in this manual, sheet 6 of 6 of **DCL-07**. 4th order Butter worth Low Pass Filter is built around U23 (IC LM324), which filters out the sampling frequency components from the Demodulation output of each modulation techniques and recovers the original signals.

- **Switch Faults**

Refer to the circuit diagram as shown later in this manual, sheet 6 of 6 of **DCL-07**. Switch fault section provides 8 switches. These switches can be used to simulate fault conditions in various parts of the circuit.

- **Audio In & Audio Out**

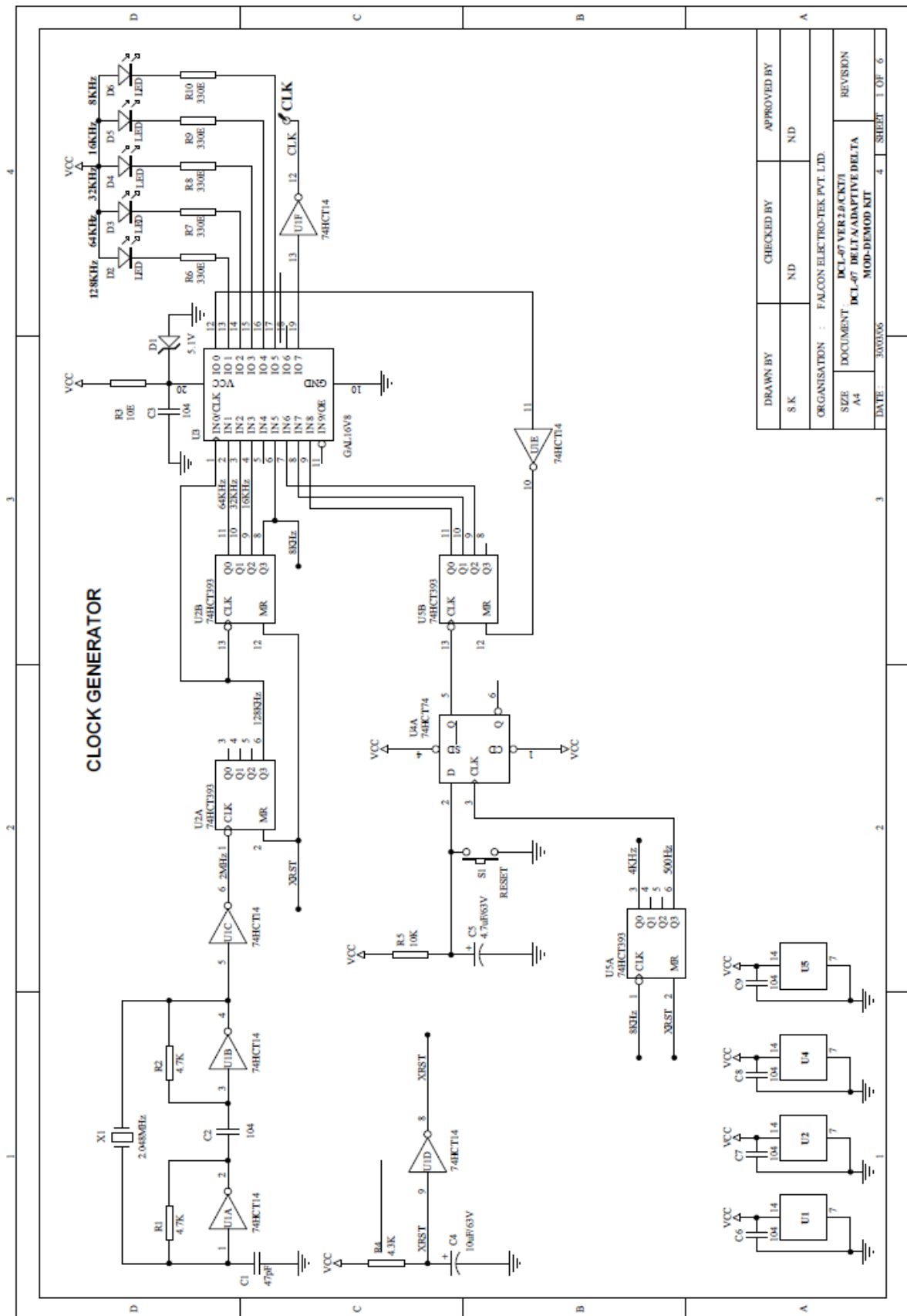
Refer to the circuit diagram as shown later in this manual, sheet 6 of 6 of **DCL-07**. These connectors are provided to connect Audio Input and Audio Output interfacing cards to DCL-07 for voice communication.

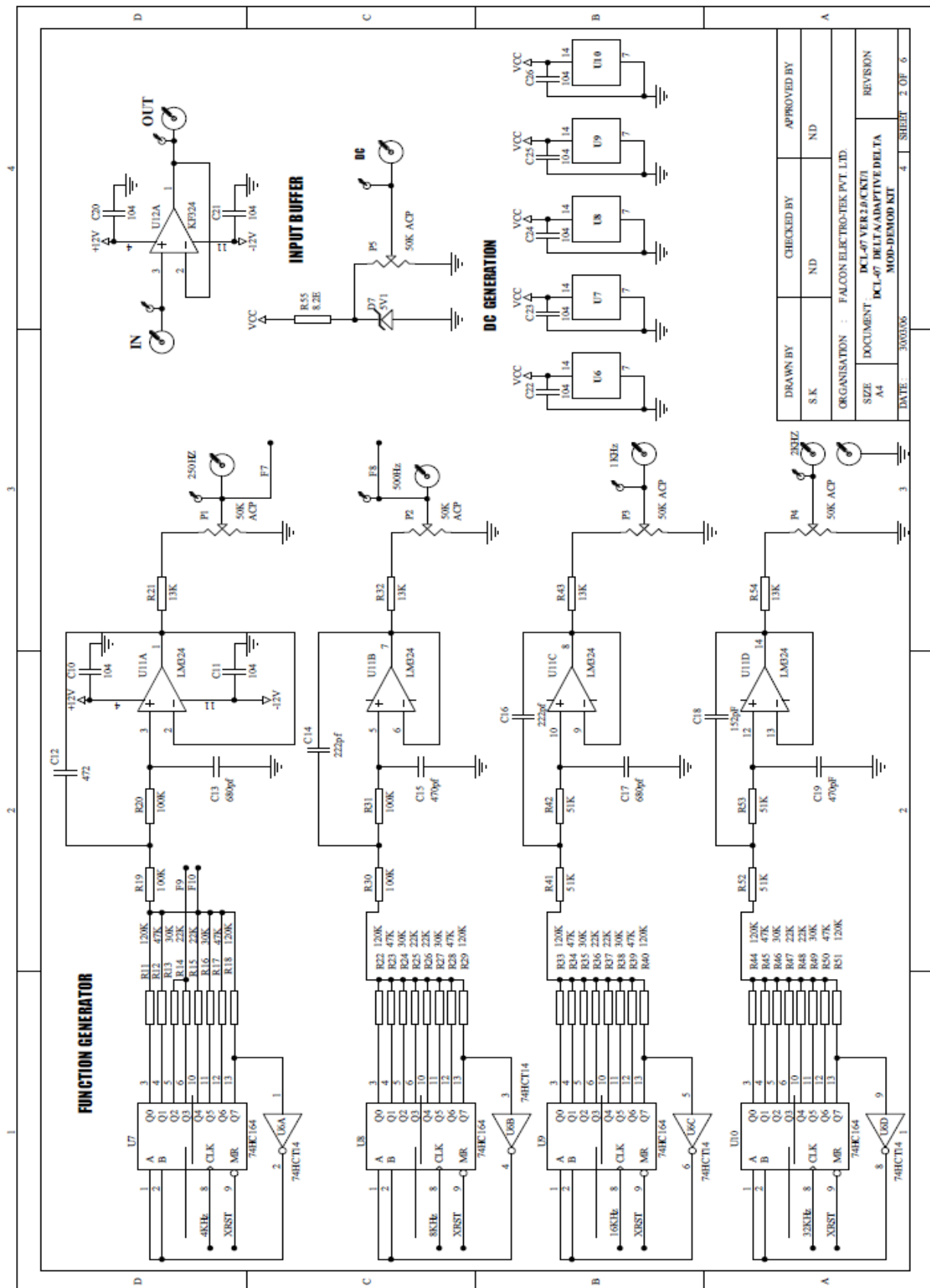
SWITCH FAULTS

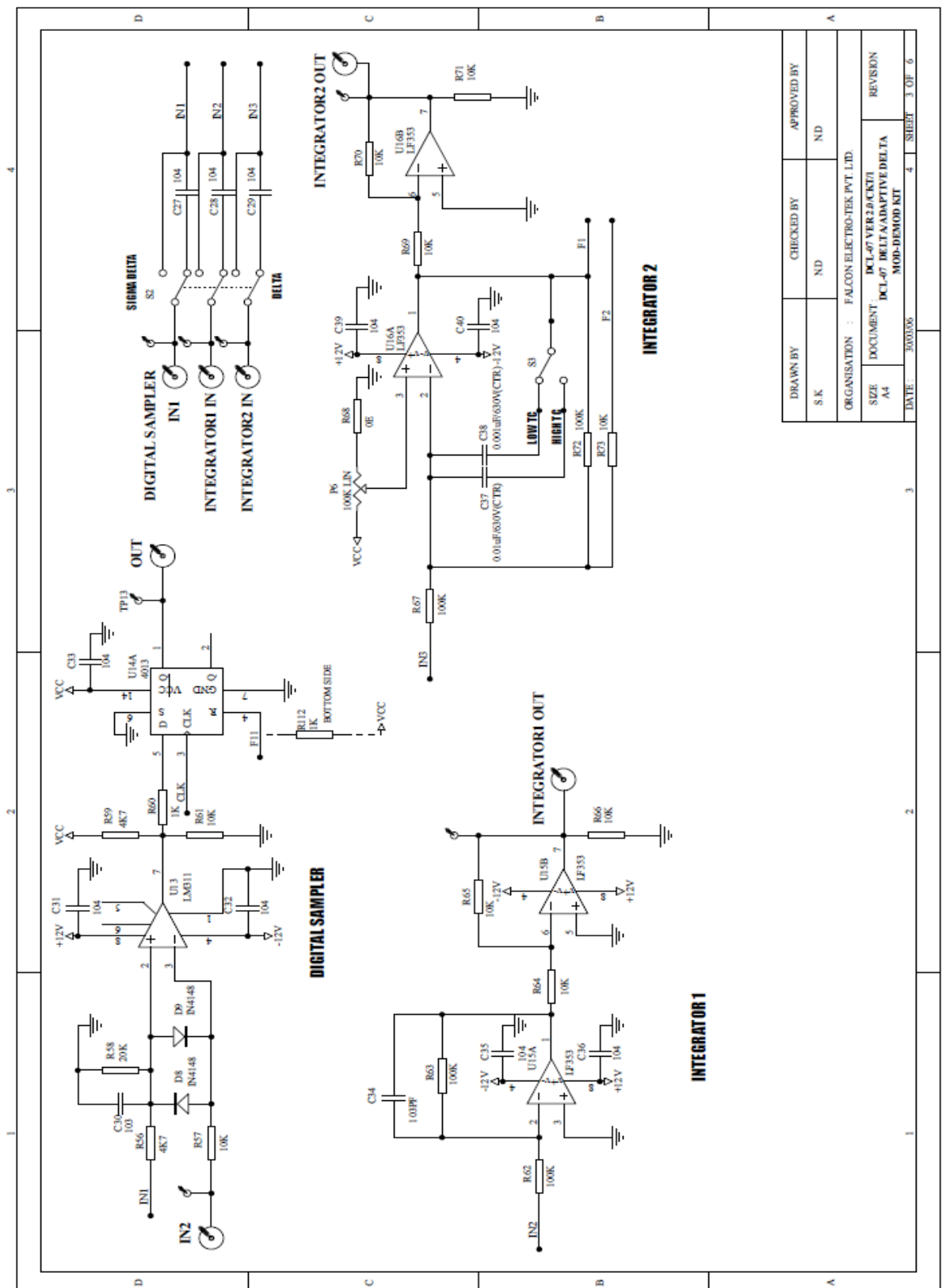
Switch fault section provides 8 switches. These switches can be used to simulate fault conditions in various parts of the circuit. The faults are normally used one at a time. To ensure that the DCL-07 kit is fully operational, all switches should be set to **OFF** position before use.

- Put switch **1** of **SF1** in Switch Fault section to **ON** position. This will short the resistor R73 (10K Ω) with R72 (100K Ω), which will reduce Integrator 2 Gain. This will affect the output of Delta / Adaptive Delta Modulation.
- Put switch **2** of **SF1** in Switch Fault section to **ON** position. This will short the resistor R80 (120K Ω) with R81 (1M Ω), which will reduce compression value. This will affect the output of compressor.
- Put switch **3** of **SF1** in Switch Fault section to **ON** position. This will short the resistor R79 (10K Ω) with R78 (100K Ω), which will reduce Integrator 3 Gain. This will affect the output of Delta / Adaptive Delta Demodulation.
- Put switch **4** of **SF1** in Switch Fault section to **ON** position. This will short the generated sine wave signals 250Hz & 500Hz, which will generate noisy sine wave.
- Put switch **5** of **SF2** in Switch Fault section to **ON** position. This will open the bypass capacitor of the 4th order low pass butter-worth filter, which results in the induction of ripples at the filter output.
- Put switch **6** of **SF2** in Switch Fault section to **ON** position. This will open R100 (1.2K Ω), which will affect internal Integrator gain of modulator IC MC3418. Results in distorted CVSD Modulator output.
- Put switch **7** of **SF2** in Switch Fault section to **ON** position. This will open the reset signal of Flip-Flop in Digital Sampler. Results in distorted output at Delta Modulator. Duration of fault gets reduced with higher clock frequency.
- Put switch **8** of **SF2** in Switch Fault section to **ON** position. This will open Resistors R14 & R15 (22K Ω), in ladder network of 250Hz sine wave signal generator. Results in distorted 250Hz sine wave signal.

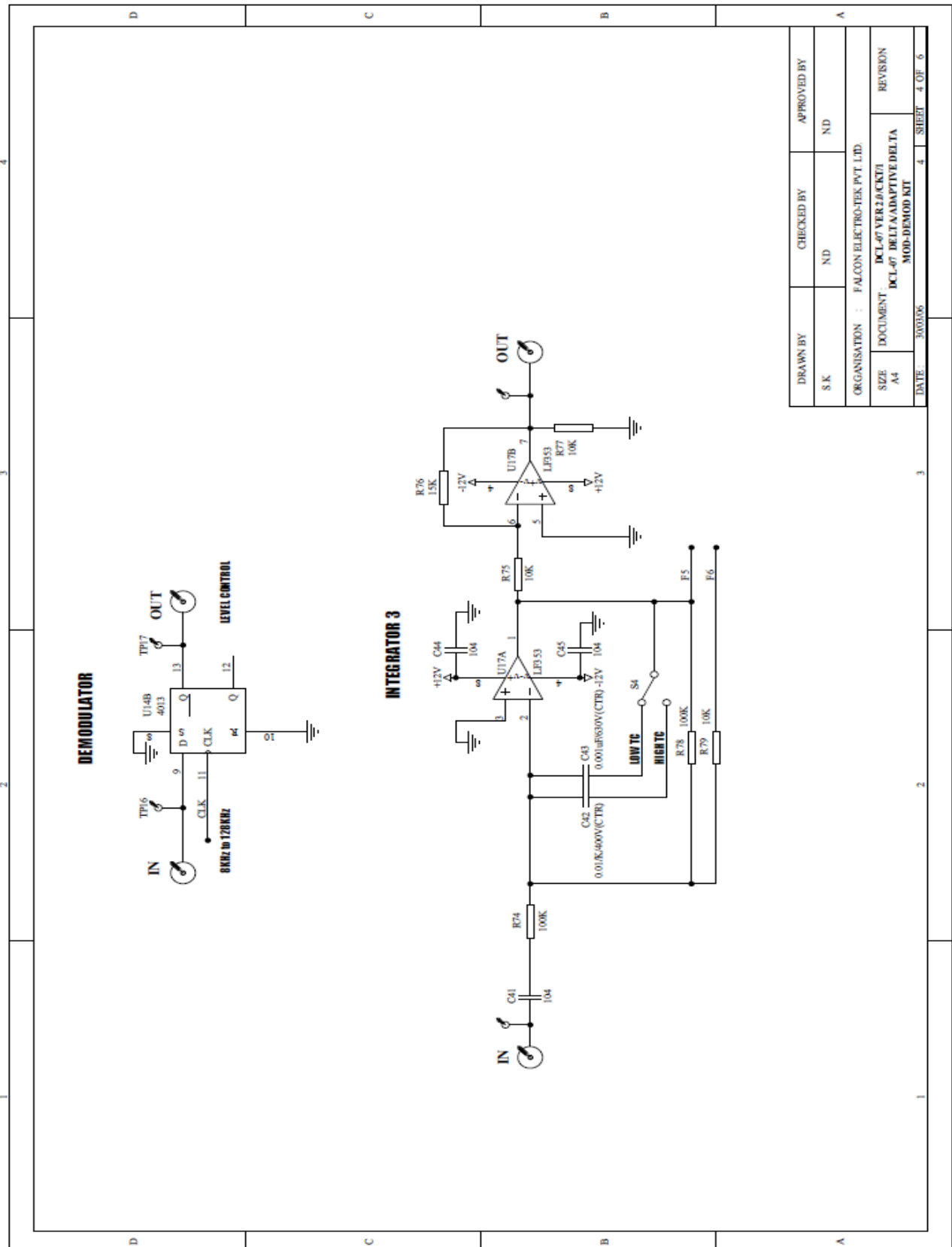
CIRCUIT DIAGRAM

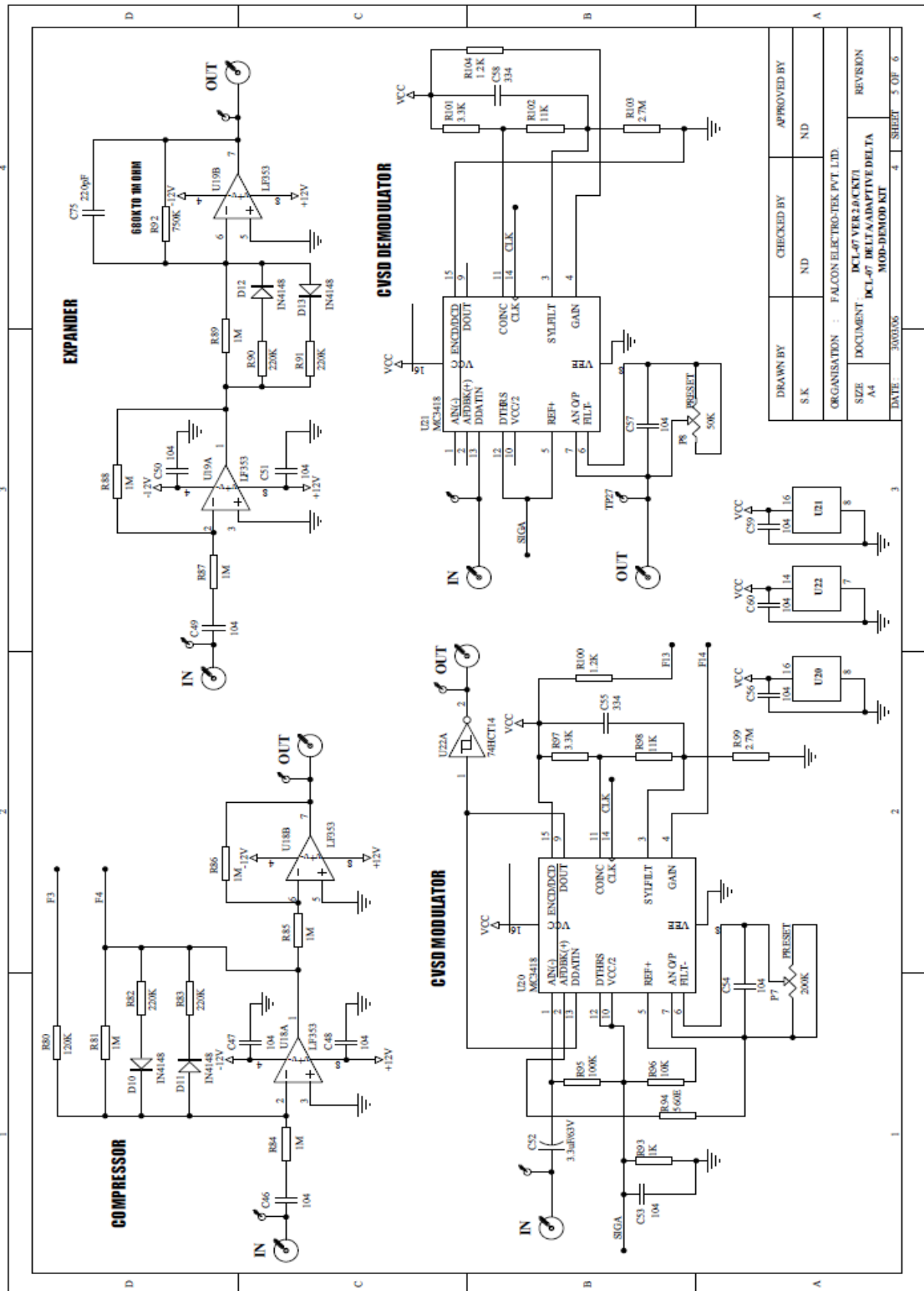


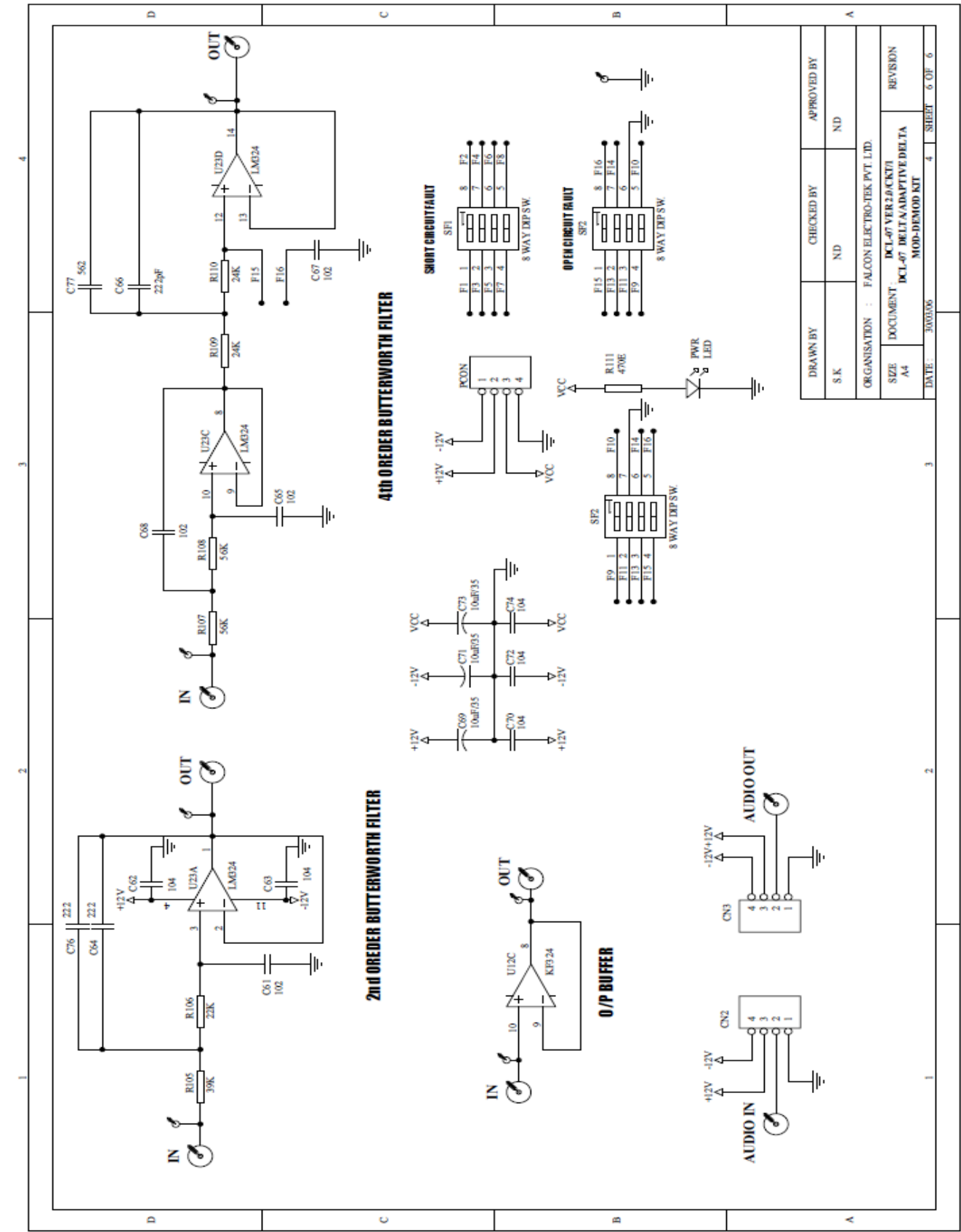




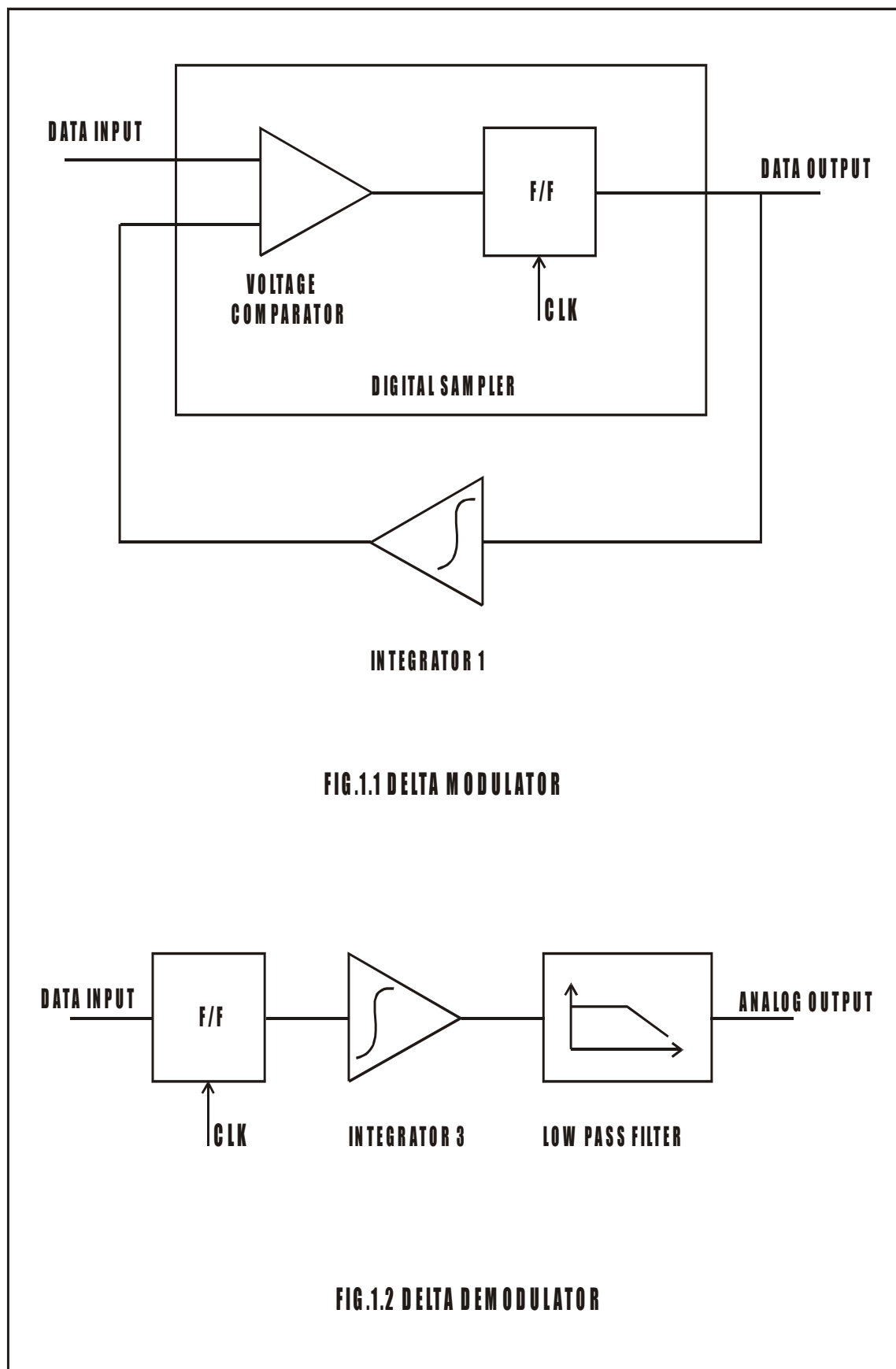
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S.K.	N.D.	N.D.
ORGANISATION	FALCON ELECTRO-TEK PVT. LTD.	
DOCUMENT	DCL-07 VER 2.0 CKT/1	REVISION
SIZE	A4	DCL-07 DELTA ADAPTIVE DELTA MOD-DEMOM KIT
DATE	30/03/06	SHEET 3 OF 6







EXPERIMENT NO. 1



EXPERIMENT NO: 1

NAME:

Delta Modulation And Demodulation

OBJECTIVE:

Study of Delta Modulation and Demodulation

THEORY:

Delta Modulation

Delta modulation is the differential pulse code modulation scheme in which the difference signal is encoded into just a single bit. In digital modulation system, the analog signal is sampled and digitally coded. This code represents the sampled amplitude of the analog signal. The digital signal is sent to the receiver through any channel in serial form. At the receiver the digital signal is decoded and filtered to get reconstructed analog signal. Sufficient number of samples is required to allow the analog signal to be reconstructed accurately. Delta modulation is a process of converting analog signal into one bit code, means only one bit is sent per sample. This bit indicates whether the signal is larger or smaller than the previous samples. The advantage of DM is that the modulator and demodulator circuits are much simpler than those used in traditional PCM.

Delta modulation is an encoding process where the logic levels of the transmitted pulses indicate whether the decoded output should rise or fall at each pulse. This is a true digital encoding process as compare to PAM, PWM and PPM.

If signal amplitude has increased in DM then modulated output is a logic level 1. If the signal amplitude has decreased the modulator output is logic level 0. Thus the output from the modulator is a series of zeroes and ones to indicate rise and fall of the waveform from the previous value.

The block diagram (Fig. 1.1) of Delta Modulation illustrates the components at the transmitter end. It consists of Digital Sampler and an Integrator at the feedback path of Digital sampler. Let assume that the base band signal $a(t)$ and its quantized approximation $i(t)$ are applied as inputs to the comparator. A comparator as its name suggests simply makes a comparison between inputs. The comparator has one fixed output $c(t)$ when $a(t) > i(t)$ and the different output when $a(t) < i(t)$ the comparator output is then latched in to a D-flip/flop which is clocked by the selected transmitter clock. Thus the output of the D-flip/flop is latched 1 or 0 synchronous with the clock edge. This binary data stream is transmitted to the receiver and is also fed to the input of integrator. The integrator output is then connected to the negative terminal of voltage comparator, thus completing the modulator circuit. The waveform of the Delta Modulator is as shown in the figure 1.5.

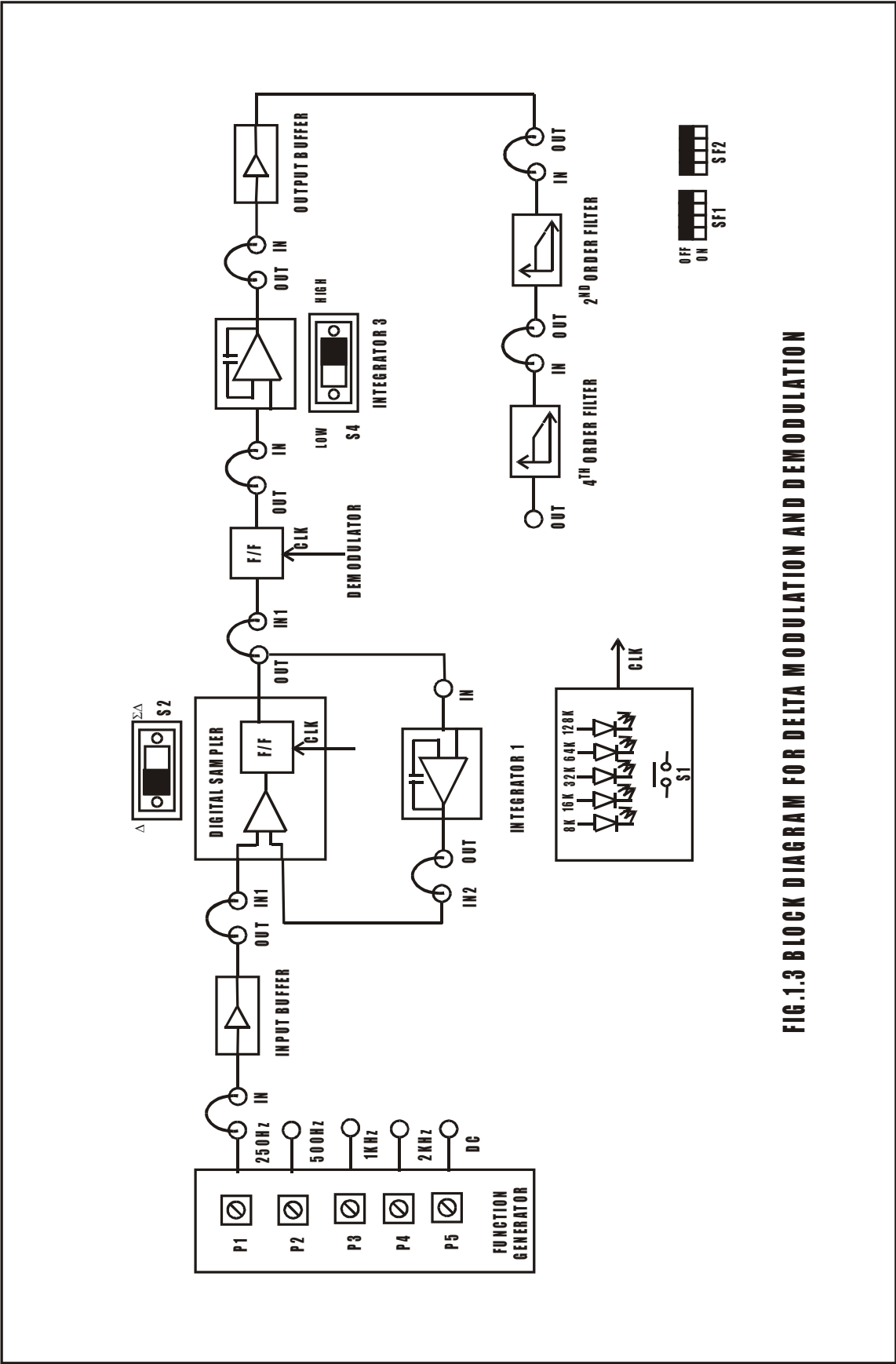


FIG.1.3 BLOCK DIAGRAM FOR DELTA MODULATION AND DEMODULATION

Delta Demodulator

The Delta Demodulator (Fig. 1.2) consists of a D-flip/flop, followed by an integrator and a 2nd and 4th order low pass butterworth filter. The Delta Demodulator receives the data stream from D-flip/flop of Delta Modulator. It latches this data at every rising edge of receiver clock. This data stream is then fed to integrator; its output tries to follow the analog signal in ramp fashion and hence is a good approximation of the signal itself. The integrator output contains sharp edges, which is smoothened out by the 2nd order, and 4th order low pass butterworth filter whose cut-off frequency is just above the audio band.

The practical use of Delta Modulation is limited due to following drawbacks:

- i) **NOISE:** A noise is defined, as any unwanted random waveform accompanying the information signal. When the signal is received at the receiver irrespective of any channel it is always accompanied by noise.
- ii) **DISTORTION:** Distortion means that the receiver output is not the true copy of the analog input signal at the transmitter. In Delta modulation, when the analog signal is greater than the integrator output the integrator ramps up to meet the analog signal. The ramping rate of integrator is constant. Therefore if the rate of change of analog input is faster than the ramping rate, the modulator is unable to catch up with the input signal. This causes a large disparity between the information signal and its quantized approximation. This error phenomenon is known as Slope over loading and causes the loss of rapidly changing information. The slope overloading waveform is as shown in the figure. The problem of slope overload can be solved by increasing the ramping rate of the integrator. But as it can be seen from the figure the effect of the large step size is to add large sharp edges at the integrator output and hence it adds to noise.
- iii) Another problem of Delta Modulation is that it is unable to pass DC information. This is not a serious limitation of the speech communication.

EQUIPMENTS:

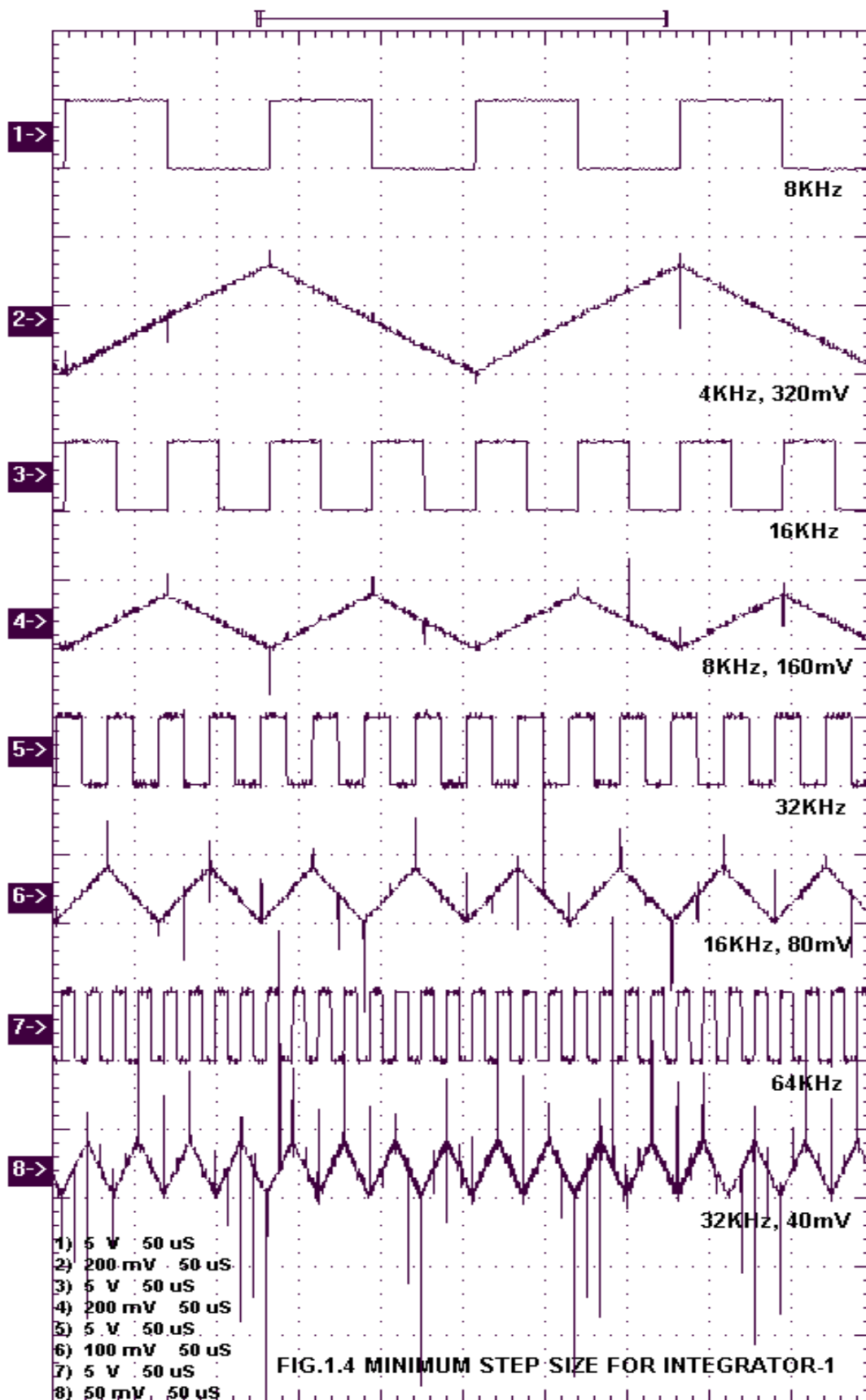
- DCL-07 kit
- Connecting chords
- Power supply
- Dual Trace Oscilloscope

NOTE: Keep All the Switch Faults (Switch SF1 & SF2) In Off Position

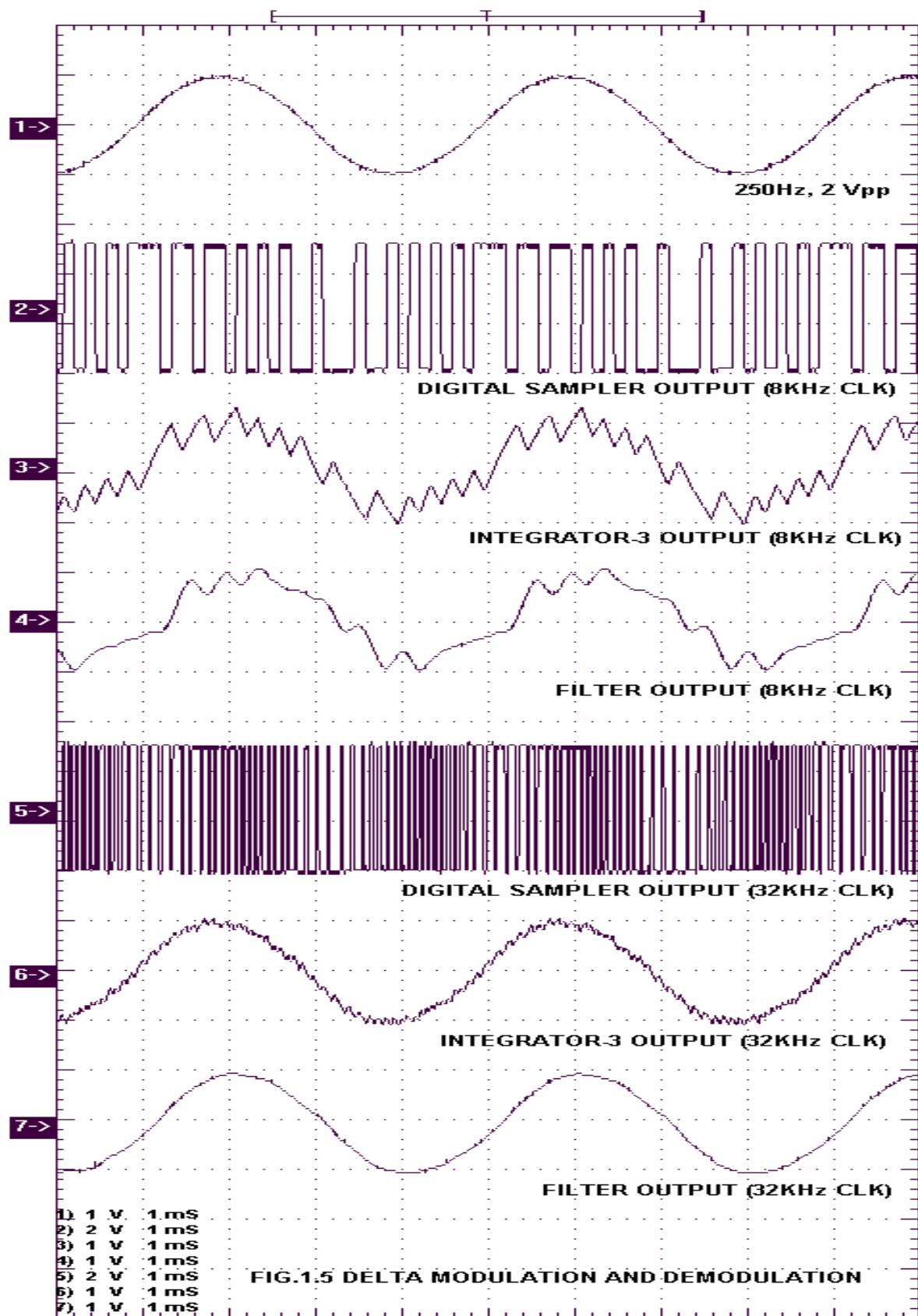
PROCEDURE:

1. Refer to the block diagram (Fig. 1.3) and carry out the following connections.
2. Connect the power supply with the proper polarity to the Kit **DCL-07** and switch it ON.
3. Select sine waves input 250Hz of 0V through pot P1 and connect post **250Hz** to post **IN** of input buffer.
4. Connect output of buffer post **OUT** to Digital Sampler input post **IN1**.
5. Then select clock rate of **8 KHz** by pressing switch **S1** selected clock is indicated by LED glow.

6. Keep Switch S2 in Δ (Delta) position.
7. Connect output of Digital Sampler post **OUT** to input post **IN** of Integrator 1.
8. Connect output of Integrator 1 post **OUT** to input post **IN2** of Digital Sampler.



9. Then observe the Delta modulated output at output of Digital Sampler post **OUT** and compare it with the clock rate selected. It is half the frequency of clock rate selected.
10. Observe the integrator output test point. It can be observe that as the clock rate is increased amplitude of triangular waveform decreases. This is called minimum step size. These waveforms are as shown in figure 1.4. Then increase the amplitude of **250Hz** sine wave upto **0.5V**. Signal approximating 250Hz is available at the integrator output. This signal is obtained by integrating the digital output resulting from Delta modulation.
11. Then go on increasing the amplitude of selected signal through the respective pot from 0 to 2V. It can be observed that the digital high makes the integrator output to go upward and digital low makes the integrator output to go downwards. Observe that the integrator output follow the input signal. The waveforms are as shown in the figure 1.5. Observe the waveforms at various test-points in the Delta modulator section.
12. Increase the amplitude of **250Hz** sine wave through pot **P1** further high and observe that the integrator output cannot follow the input signal. State the reason.
13. Repeat the above mention procedures with different signal sources and selecting the different clock rates and observe the response of Delta Modulator.
14. Connect Delta modulated output post **OUT** of Digital Sampler to the input of Delta Demodulator section post **IN** of Demodulator.
15. Connect output of Demodulator post **OUT** to the input of Integrator 3 post **IN**.
16. Connect output of Integrator 3 post **OUT** to the input of output buffer post **IN**.
17. Connect output of output buffer post **OUT** to the input of 2nd order filter post **IN**.
18. Connect output of 2nd order filter post **OUT** to the input of 4th order filter post **IN**.
19. Keep Switch S4 in **HIGH** position.
20. Then observed various tests points in Delta Demodulator section and observe the reconstructed signal through 2nd order filter and 4th order filter. Observe the waveforms as shown in figure 1.5.



SWITCH FAULTS:

Note: Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time.

- Put switch **3** of **SF1** in Switch Fault section to **ON** position. This will short the resistor R79 (10K Ω) with R78 (100K Ω), which will reduce Integrator 3 Gain. Which will affect the output of Delta / Adaptive Delta Demodulation
- Put switch **4** of **SF1** in Switch Fault section to **ON** position. This will short the generated sine wave signals 250Hz & 500Hz, which will generate noisy sine wave.
- Put switch **5** of **SF2** in Switch Fault section to **ON** position. This will open the bypass capacitor of the 4th order low pass butter-worth filter, which results in the induction of ripples at the filter output.
- Put switch **7** of **SF2** in Switch Fault section to **ON** position. This will open the reset signal of Flip-Flop in Digital Sampler. Results in distorted output at Delta Modulator. Duration of fault gets reduced with higher clock frequency.
- Put switch **8** of **SF2** in Switch Fault section to **ON** position. This will open Resistors R14 & R15 (22K Ω), in ladder network of 250Hz sine wave signal generator. Results in distorted 250Hz sine wave signal.

OBSERVATION:

Observe the following signal on oscilloscope and plot it on the paper. (Fig. 1.4 & Fig.1.5)

Sampling clock

Input Signal

Integrator 1 output at feedback loop for Delta modulator

Digital sampler Output

Demodulator Output

Integrator 3 output

Filter Outputs

EXPERIMENT NO. 2

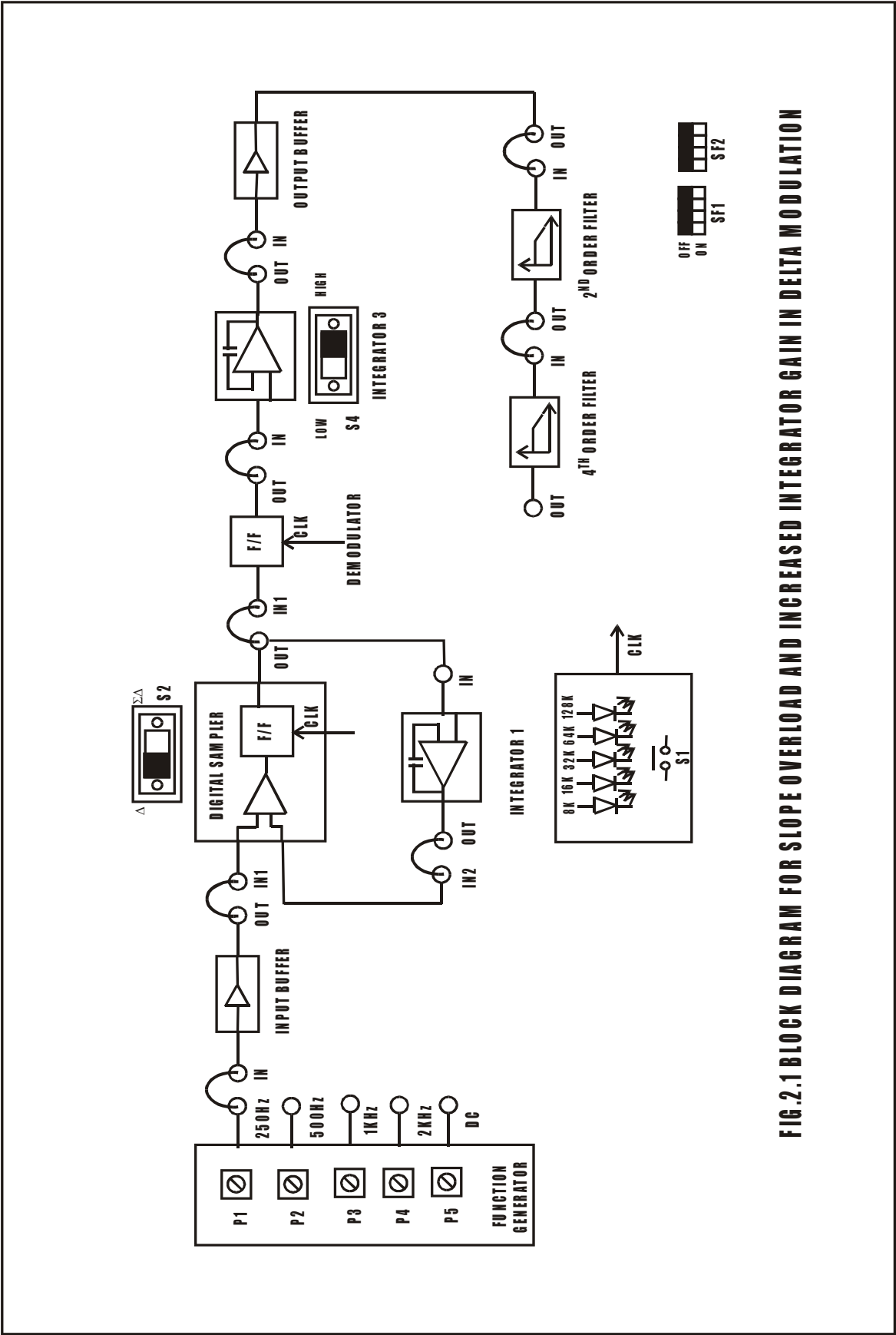


FIG.2.1 BLOCK DIAGRAM FOR SLOPE OVERLOAD AND INCREASED INTEGRATOR GAIN IN DELTA MODULATION

EXPERIMENT NO: 2

NAME:

Slope Overload And Increased Integrator Gain.

OBJECTIVE:

Study of slope overload and increased integrator gain in Delta Modulation

THEORY:

Slope overload occurs when the slope of the input signal exceeds the delta modulators maximum slope. The slope overload can be reduced by increasing the size of the step increment. Slope overload is one of the fundamental limitations of the Delta Modulation. This problem is solved in Adaptive Delta Modulation.

As it has been seen, when the analog signal is greater than the integrator output the integrator ramps up to meet the analog signal. The ramping rate of integrator is constant. Therefore if the rate of change of analog input is faster than the ramping rate, the modulator is unable to catch up with the input signal. This causes a large disparity between the information signal and its quantized approximation. This error phenomenon is known as Slope over loading and causes the loss of rapidly changing information. The slope overloading waveform is as shown in figure 2.2. The problem of slope overloading can be solved by increasing the ramping rate of the integrator. But as it can be seen from the figure the effect of the large step size is to add large sharp edges at the integrator output and hence it adds to noise. Time constant of the integrator variable from 0.5ms to 1.5ms for testing purpose

EQUIPMENTS:

- DCL-07 kit
- Connecting chords
- Power supply
- Dual Trace Oscilloscope

NOTE: Keep All the Switch Faults (Switch SF1 & SF2) In Off Position

PROCEDURE:

1. Refer to the block diagram (Fig. 2.1) and carry out the following connections.
2. Connect the power supply with the proper polarity to the Kit **DCL-07** and switch it ON.
3. Select sine waves input 250Hz of 4V through pot P1 and connect post **250Hz** to post **IN** of input buffer.
4. Connect output of buffer post **OUT** to Digital Sampler input post **IN1**.

5. Then select clock rate of **8 KHz** by pressing switch **S1** selected clock is indicated by LED glow.
6. Keep Switch S2 in Δ (Delta) position.
7. Connect output of Digital Sampler post **OUT** to input post **IN** of Integrator 1.
8. Connect output of Integrator 1 post **OUT** to input post **IN2** of Digital Sampler.
9. Connect Delta modulated output post **OUT** of Digital Sampler to the input of Delta Demodulator section post **IN** of Demodulator.
10. Connect output of Demodulator post **OUT** to the input of Integrator 3 post **IN**.
11. Connect output of Integrator 3 post **OUT** to the input of output buffer post **IN**.
12. Connect output of output buffer post **OUT** to the input of 2nd order filter post **IN**.
13. Connect output of 2nd order filter post **OUT** to the input of 4th order filter post **IN**.
14. Keep Switch S4 in **HIGH** position.
15. Observe the integrator output at different clock rates.
16. Then perform the Experiment by changing sine wave frequency step by step from 500Hz to 2KHz. Select **1KHz** sine wave of **4V** amplitude as input signal you will observe slope overload effect as shown in figure 2.2. Note that as the frequency of the analog signal increases there is less conversion of 1's and 0's in Delta modulated o/p than low frequency and more information loss is there due to slope over load and we get reduced amplitude at filter output. The increase in frequency makes the signal vary so quickly that the integrator output cannot ramp fast enough to catch up with it, and result is known as slope overloading. Observe the waveforms as shown in figure 2.2.
17. Slope overloading can be overcome by either reducing the maximum input frequency (Here 250Hz) of the Delta Modulator and reducing the maximum input amplitude (less than 2V) or increasing the integrator gain for Delta Modulator and for Delta Demodulator.

SWITCH FAULTS:

Note: Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time.

- Put switch **3** of **SF1** in Switch Fault section to **ON** position. This will short the resistor R79 (10K Ω) with R78 (100K Ω), which will reduce Integrator 3 Gain. This will affect the output of Delta / Adaptive Delta Demodulation.
- Put switch **4** of **SF1** in Switch Fault section to **ON** position. This will short the generated sine wave signals 250Hz & 500Hz, which will generate noisy sine wave.
- Put switch **5** of **SF2** in Switch Fault section to **ON** position. This will open the bypass capacitor of the 4th order low pass butter-worth filter, which results in the induction of ripples at the filter output.
- Put switch **7** of **SF2** in Switch Fault section to **ON** position. This will open the reset signal of Flip-Flop in Digital Sampler. Results in distorted output at Delta Modulator. Duration of fault gets reduced with higher clock frequency.
- Put switch **8** of **SF2** in Switch Fault section to **ON** position. This will open Resistors R14 & R15 (22K Ω), in ladder network of 250Hz sine wave signal generator. Results in distorted 250Hz sine wave signal.

OBSERVATION:

Observe the following signal on oscilloscope and plot it on the paper. (Fig. 2.2)

Sampling clock

Input Signal

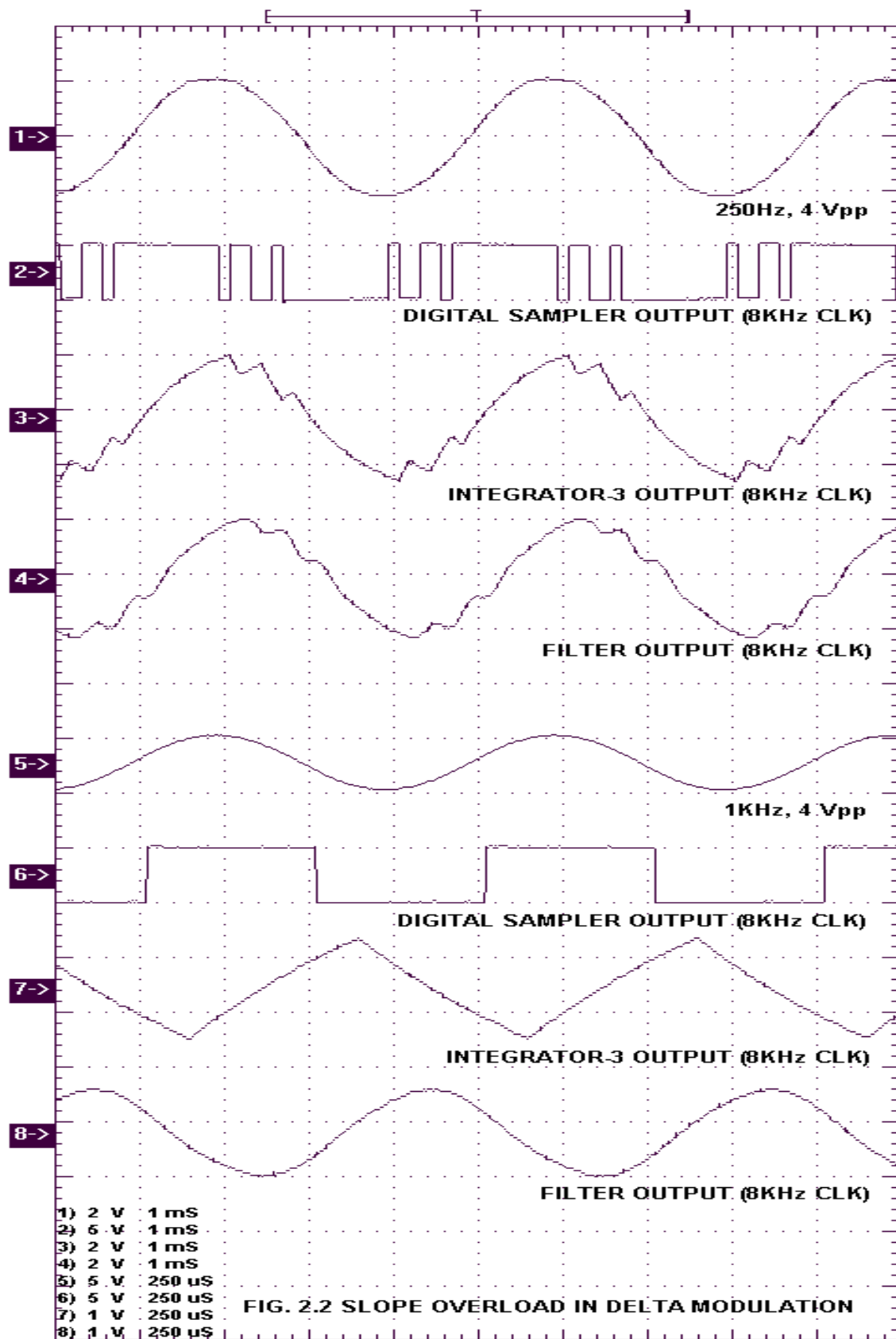
Integrator 1 output at feedback loop for Delta modulator

Digital sampler Output

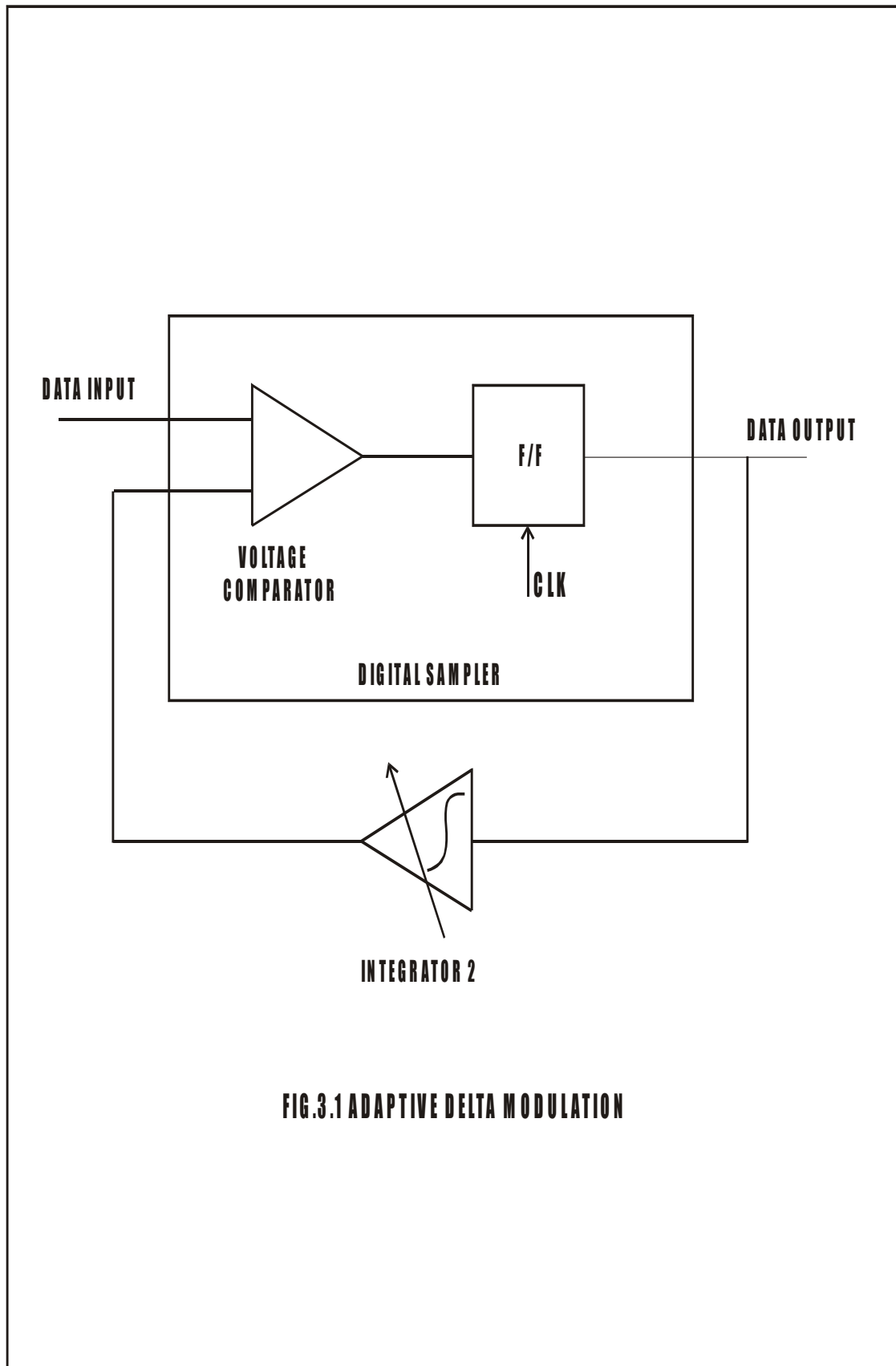
Demodulator Output

Integrator 3 output

Filter Outputs.



EXPERIMENT NO. 3



EXPERIMENT NO: 3

NAME:

Adaptive Delta Modulation And CVSD

OBJECTIVE:

Study of Adaptive Delta Modulation and Demodulation with Continuously Variable Slope Delta Modulator (CVSD)

THEORY:

As seen in earlier system Delta Modulation system is unable to chase the rapidly changing information of the analog signal, which gives rise to distortion and poor quality reception. The problem can be overcome by increasing the integrator gain. Adaptive Delta Modulation is a variation of Delta Modulation, which offers relief from disadvantage of DM by adopting the step size to accommodate changing signal conditions. If the input signal is large, step is caused to increase, thereby reducing slope overload effects. The block diagram of ADM is as shown in fig 3.1. It is same as Delta Modulation except the variable gain circuit and step size controller. The controller keeps sensing the slope condition of the message conveyed. If the slope is large the controller output causes the variable gain circuit to have large gain. If the slope is small, the controller output causes a small gain. In certain cases Adaptive Delta Modulation do not change step size on a pulse-to-pulse basis, but changes are made much more slowly, such slow control is referred to as syllabic. The usual implementation involves a continuously variable slope Delta (CVSD). There are varieties of IC's for CVSD encoding and decoding in today's semiconductor market.

The CVSD is the simple alternative to more complex conventional conversion techniques in system requiring digital communication of analog signals. The CVSD A/D is well suited for the requirements of digital communications. A Delta Modulator consists of a comparator in the forward path and an integrator in the feedback path of a simple control loop. The input to the comparator is the simple analog signal and the integrator output. The comparator output is the difference between the input voltage and the integrator output. That sign bit is the digital output and also control the direction of ramp in the integrator. The output of comparator is fed to the sampler. Then the sampler output is fed to the slope polarity switch and level detect algorithm. The level detect algorithm is again fed to the slope magnitude control followed by slope polarity switch. The output slope polarity switch is fed to the integrator in the control loop. With no input at the transmitter a continuous 1 and 0 alternations are transmitted. The outstanding characteristic is its ability to transmit the intelligible voice out at relatively low data rate. Companded PCM for telephone quality transmission requires about 64Kbits/sec. data rate/channel. CVSD produces equal quality at 32Kbit/sec. In CVSD Decoder CVSD mod output is fed to the input of comparator.

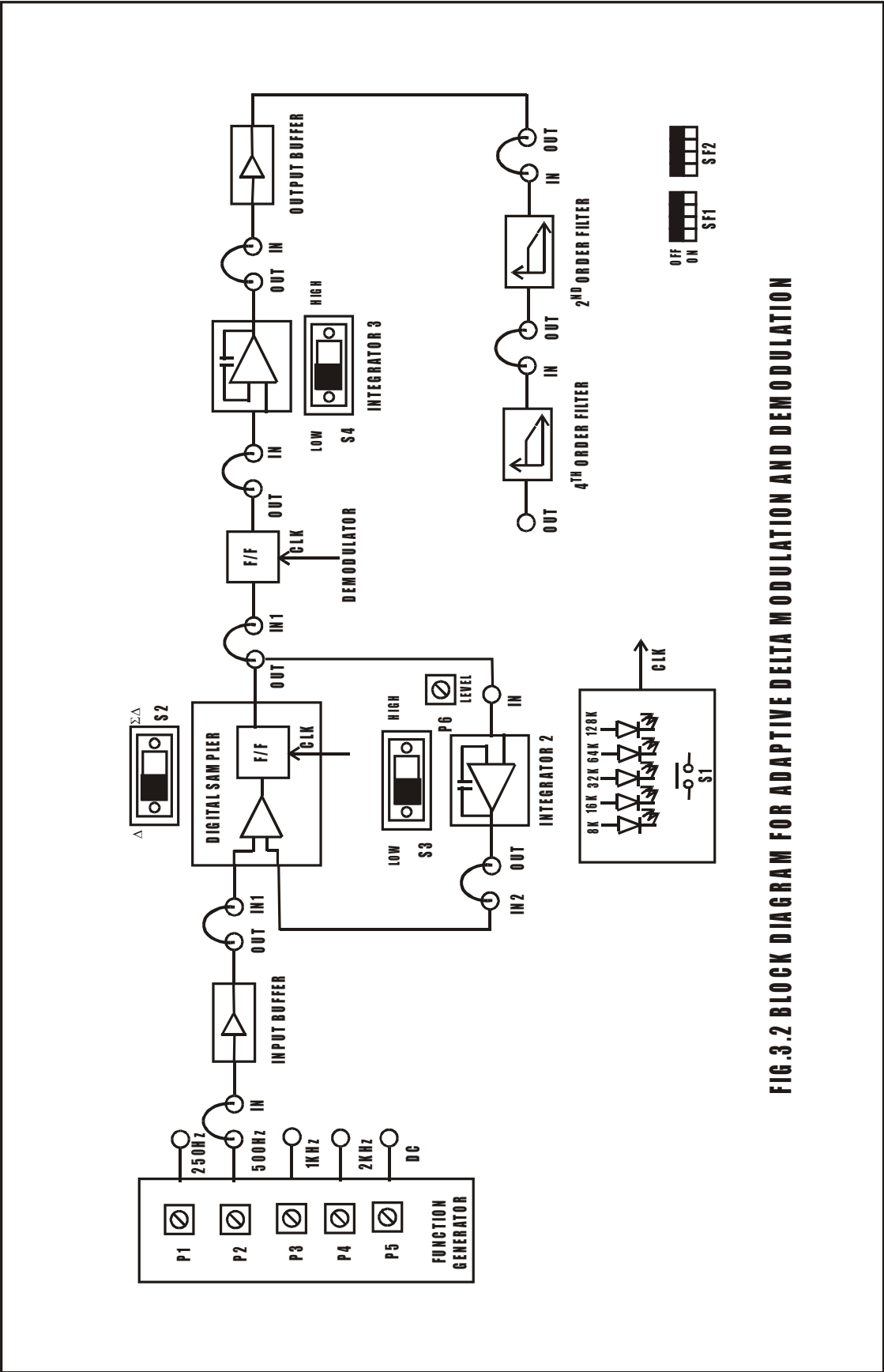


FIG.3.2 BLOCK DIAGRAM FOR ADAPTIVE DELTA MODULATION AND DEMODULATION

The comparator output is fed to the internal shift register. Then the output of internal shift register is fed to the digital logic followed by slope polarity switch and integrator. The output of integrator is fed to the low pass filters for the reconstruction of original signal.

EQUIPMENTS:

- DCL –07 kit
- Connecting chords
- Power supply
- Dual Trace Oscilloscope

NOTE: Keep All the Switch Faults (Switch SF1 & SF2) In Off Position

PROCEDURE:

Adaptive Delta Modulation

1. Refer to the block diagram (Fig. 3.2) and carry out the following connections.
2. Connect the power supply with the proper polarity to the Kit **DCL-07** and switch it ON.
3. Select sine wave input **500Hz** of **2V** or above through pot **P2** and connect post **1 KHz** to post **IN** of input buffer.
4. Connect output of buffer post **OUT** to Digital Sampler input post **IN1**.
5. Then select clock rate of **32 KHz** by pressing switch **S1** selected clock is indicated by LED glow.
6. Keep Switch S2 in Δ (Delta) position.
7. Connect output of Digital Sampler post **OUT** to input post **IN** of Integrator 2.
8. Connect output of Integrator 2 post **OUT** to input post **IN2** of Digital Sampler.
9. Keep Switch S3 in **LOW** position.
10. Connect Delta modulated output post **OUT** of Digital Sampler to the input of Delta Demodulator section post **IN** of Demodulator.
11. Connect output of Demodulator post **OUT** to the input of Integrator 3 post **IN**.
12. Connect output of Integrator 3 post **OUT** to the input of output buffer post **IN**.
13. Connect output of output buffer post **OUT** to the input of 2nd order filter post **IN**.
14. Connect output of 2nd order filter post **OUT** to the input of 4th order filter post **IN**.
15. Keep Switch S4 in **LOW** position.
16. Observe the output of filter at post **OUT** of 4th order filter.
17. Repeat the above mention procedures with different signal sources and selecting the different clock rates and observe the response of Adaptive delta modulation. This follows input signal even if amplitude & frequency of input signal increases. Adaptive delta modulator matches the slope of the input signal due to low time constant.
18. We will get better recovery of signal as we go on increasing the clock rates (e.g. 128 KHz).

SWITCH FAULTS:

Note: Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time. Put switch 1 of SF1 in Switch Fault section to ON position. This will short the resistor R73 (10K Ω) with R72 (100K Ω), which will reduce Integrator 2 Gain. Which will affect the output of Delta / Adaptive Delta Modulation

- Put switch **3** of **SF1** in Switch Fault section to **ON** position. This will short the resistor R79 (10K Ω) with R78 (100K Ω), which will reduce Integrator 3 Gain. This will affect the output of Delta / Adaptive Delta Demodulation.
- Put switch **4** of **SF1** in Switch Fault section to **ON** position. This will short the generated sine wave signals 250Hz & 500Hz, which will generate noisy Sine wave
- Put switch **5** of **SF2** in Switch Fault section to **ON** position. This will open the bypass capacitor of the 4th order low pass butter-worth filter, which results in the induction of ripples at the filter output.
- Put switch **7** of **SF2** in Switch Fault section to **ON** position. This will open the reset signal of Flip-Flop in Digital Sampler. Results in distorted output at Delta Modulator. Duration of fault gets reduced with higher clock frequency.
- Put switch **8** of **SF2** in Switch Fault section to **ON** position. This will open Resistors R14 & R15 (22K Ω), in ladder network of 250Hz sine wave signal generator. Results in distorted 250Hz sine wave signal.

OBSERVATION:

Observe the following signal on oscilloscope and plot it on the paper

Sampling clock

Input Signal

Integrator 2 output at feedback loop for Adaptive Delta modulator

Digital sampler Output

Demodulator Output

Integrator 3 output

Filter Outputs.

NOTE: Keep All “the Switch Faults (Switch Sf1 & Sf2) In Off Position

CVSD

- Refer to the block diagram (Fig. 3.3) and carry out the following connections.
- Connect the power supply with the proper polarity to the Kit **DCL-07** and switch it ON.

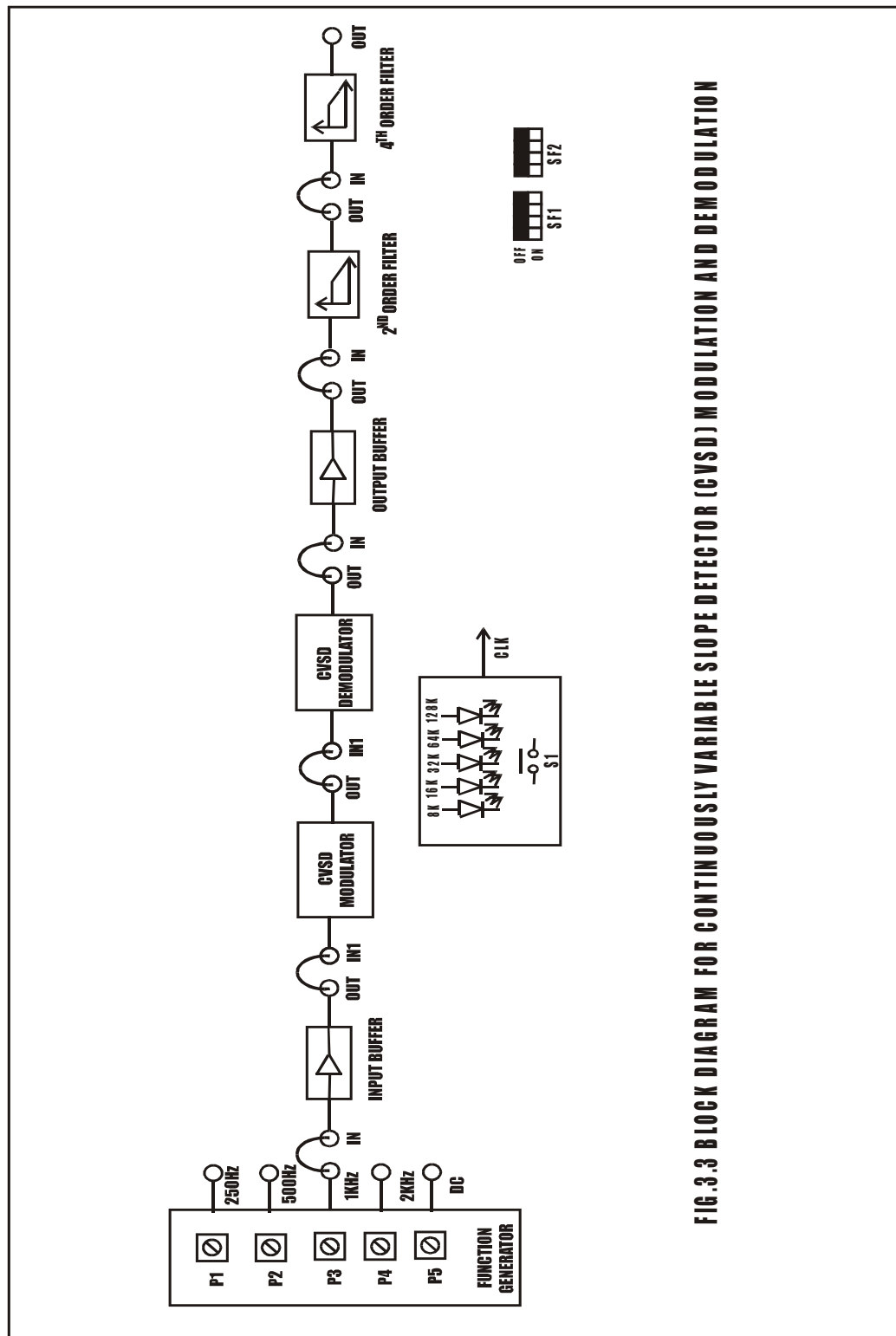


FIG.3.3 BLOCK DIAGRAM FOR CONTINUOUSLY VARIABLE SLOPE DETECTOR (CVSD) MODULATION AND DEMODULATION

- Select sine wave input **1 KHz** of **2V** or above through pot **P3** and connect post **1 KHz** to post **IN** of input buffer.
- Connect output of buffer post **OUT** to the **CVSD** modulator input post **IN**.
- Select the clock rate of **32 KHz** by pressing the **S1**
- Observe the **CVSD** output.
- Connect the output of **CVSD** modulator post **OUT** to the input of **CVSD** Demodulator post **IN**. Observe the waveforms of the **CVSD** as shown in figure 3.5.
- Connect **CVSD** Demodulated output post **OUT** to the input of output buffer post **IN**.
- Connect output of output buffer post **OUT** to the input of 2nd order filter post **IN**.
- Connect output of 2nd order filter post **OUT** to the input of 4th order filter post **IN**.
- Repeat the above mention procedures with different signal sources and selecting the different clock rates and observe the response of CVSD Modulator and Demodulator.
- Observe all signals on dual channel through oscilloscope probes. We will get clear reception at high clock rates (e.g. 128 KHz).

SWITCH FAULTS:

Note: Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time.

- Put switch **4** of **SF1** in Switch Fault section to **ON** position. This will short the generated sine wave signals 250Hz & 500Hz, which will generate noisy sine wave.
- Put switch **5** of **SF2** in Switch Fault section to **ON** position. This will open the bypass capacitor of the 4th order low pass butter-worth filter, which results in the induction of ripples at the filter output.
- Put switch **6** of **SF2** in Switch Fault section to **ON** position. This will open R100 (1.2K Ω), which will affect internal Integrator gain of modulator IC MC3418. Results in distorted CVSD Modulator output.
- Put switch **8** of **SF2** in Switch Fault section to **ON** position. This will open Resistors R14 & R15 (22K Ω), in ladder network of 250Hz sine wave signal generator. Results in distorted 250Hz sine wave signal.

OBSERVATION:

Observe the following signal on oscilloscope and plot it on the paper. (Fig. 3.5)

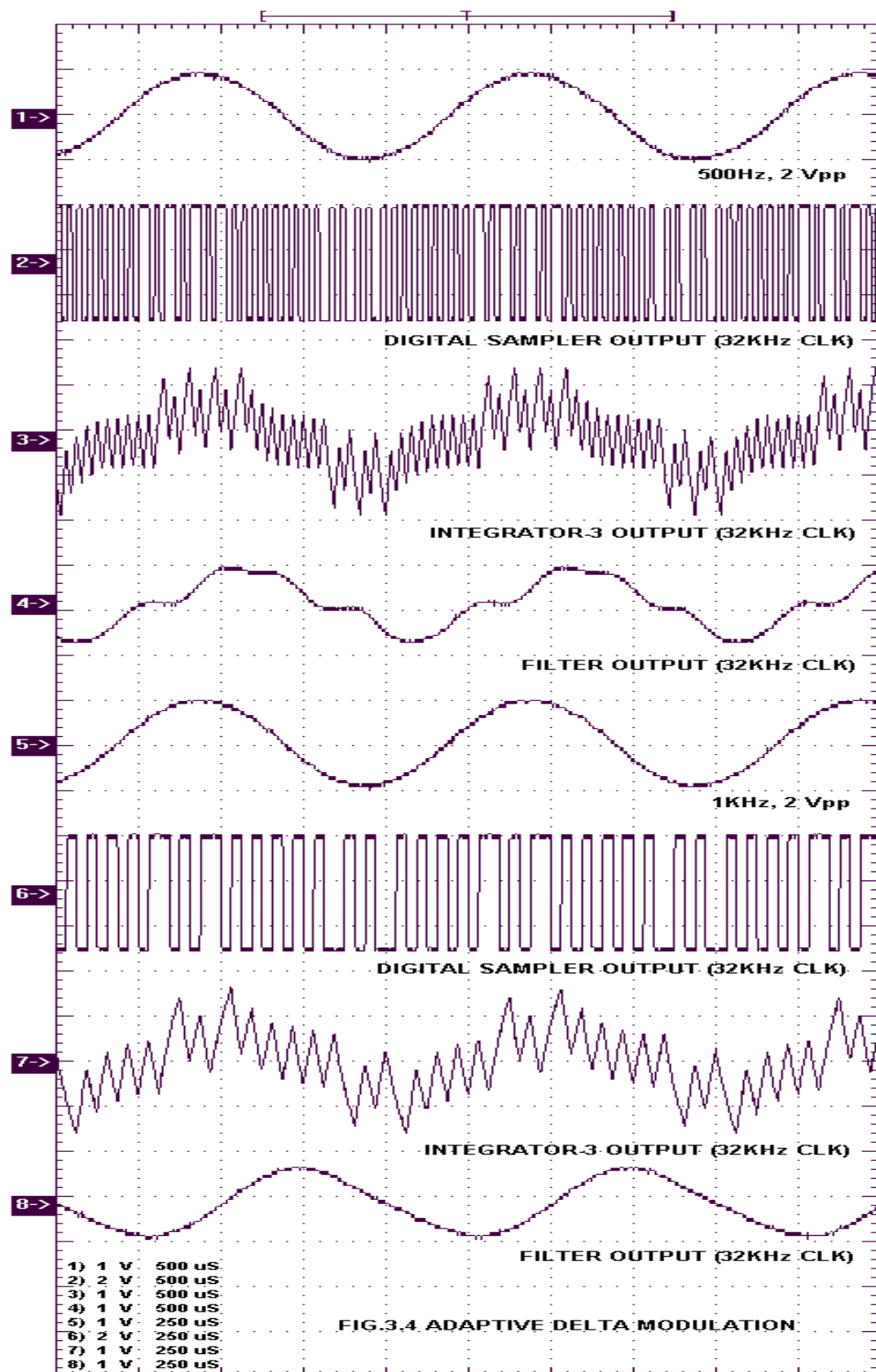
Sampling clock

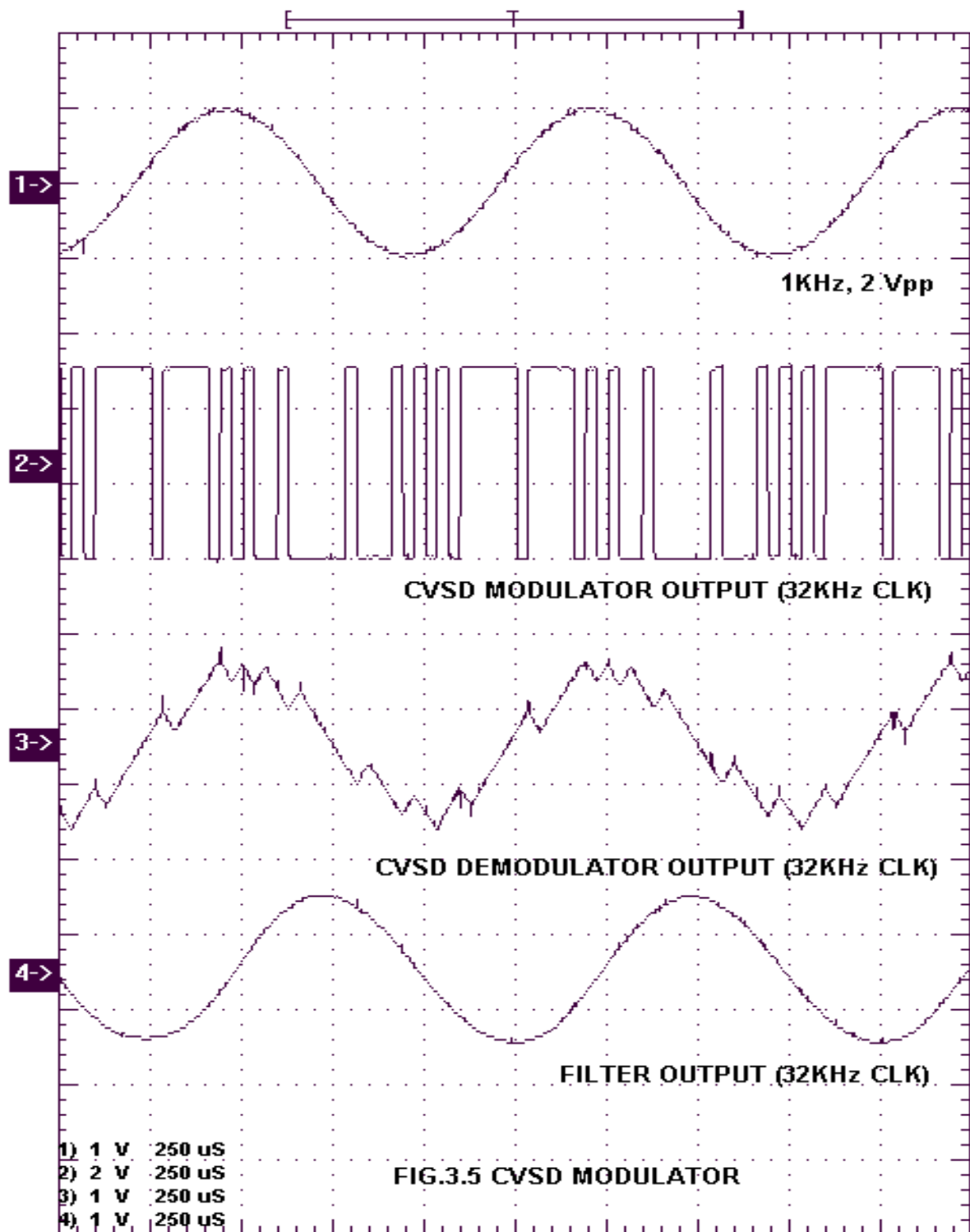
Input Signal

CVSD Modulator Output

CVSD Demodulator Output

Filter Outputs.





EXPERIMENT NO. 4

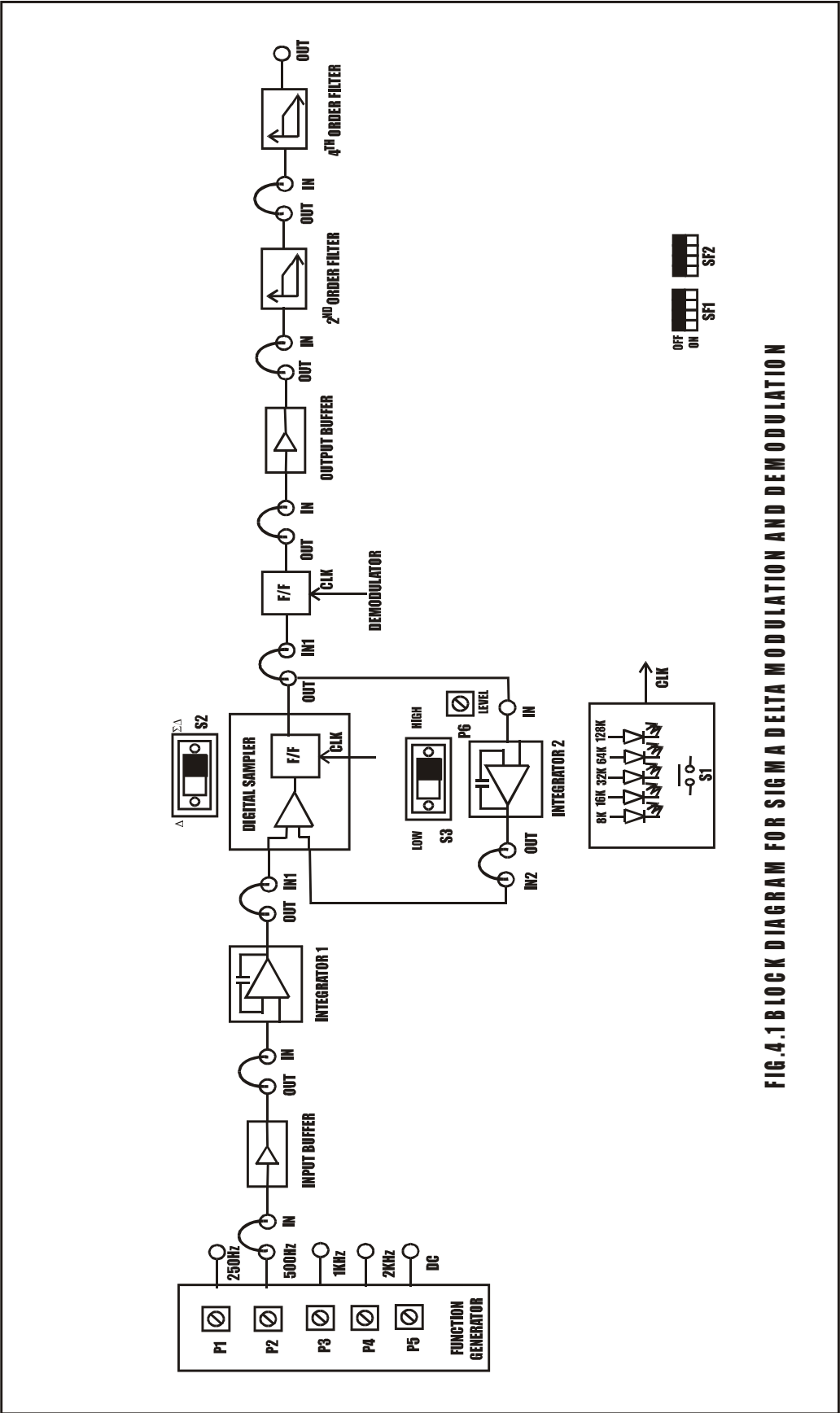


FIG.4.1 BLOCK DIAGRAM FOR SIGMA DELTA MODULATION AND DEMODULATION

EXPERIMENT NO: 4

NAME:

Sigma Delta Modulation And Demodulation

OBJECTIVE:

Study of Sigma Delta Modulation And Demodulation

THEORY:

The quantizer (feedback) input in Delta Modulation is viewed as an approximation to the derivative of the incoming signal. This behavior leads to a drawback of delta modulation in that transmission disturbances such as noise result in an accumulative error in demodulated signal. This drawback is overcome by integrating the incoming signal prior to delta modulation.

Correlation between adjacent samples of the delta modulator input is increased, which tends to improve overall system performance by reducing the variances of the error signal at the feedback input. This way modulator now responds to the actual voltage levels rather than the change in voltage level. Integrator at input of Delta modulator is called Sigma Delta modulator. Through Sigma delta modulator we can transmit DC signal and Receiver design becomes simple. We require only low pass filter at the receiver side to demodulate the modulated signal and get back original information signal.

EQUIPMENTS:

- DCL-07 kit.
- Connecting chords
- Power supply
- Dual Trace Oscilloscope

NOTE: Keep All the Switch Faults (Switch SF1 & SF2) In Off Position

PROCEDURE:

1. Refer to the block diagram (Fig. 4.1) and carry out the following connections.
2. Connect the power supply with the proper polarity to the Kit **DCL-07** and switch it ON.
3. Select sine wave input **500Hz** of **2V** or above through pot **P2** and connect post **500Hz** to post **IN** of input buffer.
4. Connect output of buffer post **OUT** to input post **IN** of Integrator 1.
5. Connect output of Integrator 1 post **OUT** to Digital Sampler input post **IN1**.
6. Then select clock rate of **16 KHz** by pressing switch **S1** selected clock is indicated by LED glow.
7. Keep Switch S2 in $\Sigma\Delta$ (Sigma Delta) position.
8. Connect output of Digital Sampler post **OUT** to input post **IN** of Integrator 2.

9. Connect output of Integrator 2 post **OUT** to input post **IN2** of Digital Sampler.
10. Connect Delta modulated output post **OUT** of Digital Sampler to the input of Demodulator post **IN**.
11. Connect output of Demodulator post **OUT** to the input of Output Buffer post **IN**.
12. Connect output of Output Buffer post **OUT** to the input of 2nd order filter post **IN**.
13. Connect output of 2nd order filter post **OUT** to the input of 4th order filter post **IN**.
14. Keep Switch S3 in **HIGH** position.
15. Connect oscilloscope probe at Digital Sampler output, By using level control pot P6 change input level of integrator-2 to obtain output similar to the one shown in waveforms fig.4.2 (Get output in symmetry at both positive peak and negative peak).
16. Then perform the Experiment by changing sine wave frequency step by step from 500Hz to 2KHz and by applying DC signal also.
17. For DC input signal apply clock 64KHz.
18. Observe all signals on dual channel through oscilloscope probes.
19. Repeat the above mention procedures by increasing the clock frequency to get better reproduction of the input signal.

SWITCH FAULTS:

Note: Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time.

- Put switch **1** of **SF1** in Switch Fault section to **ON** position. This will short the resistor R73 (10K Ω) with R72 (100K Ω), which will reduce Integrator 2 Gain. This will affect the output of Delta / Adaptive Delta Modulation.
- Put switch **4** of **SF1** in Switch Fault section to **ON** position. This will short the generated sine wave signals 250Hz & 500Hz, which will generate noisy sine wave.
- Put switch **5** of **SF2** in Switch Fault section to **ON** position. This will open the bypass capacitor of the 4th order low pass butter-worth filter, which results in the induction of ripples at the filter output.
- Put switch **7** of **SF2** in Switch Fault section to **ON** position. This will open the reset signal of Flip-Flop in Digital Sampler. Results in distorted output at Delta Modulator. Duration of fault gets reduced with higher clock frequency.
- Put switch **8** of **SF2** in Switch Fault section to **ON** position. This will open Resistors R14 & R15 (22K Ω), in ladder network of 250Hz sine wave signal generator. Results in distorted 250Hz sine wave signal.

OBSERVATION:

Observe the following signal on oscilloscope and plot it on the paper. (Fig. 4.2)

Sampling clock

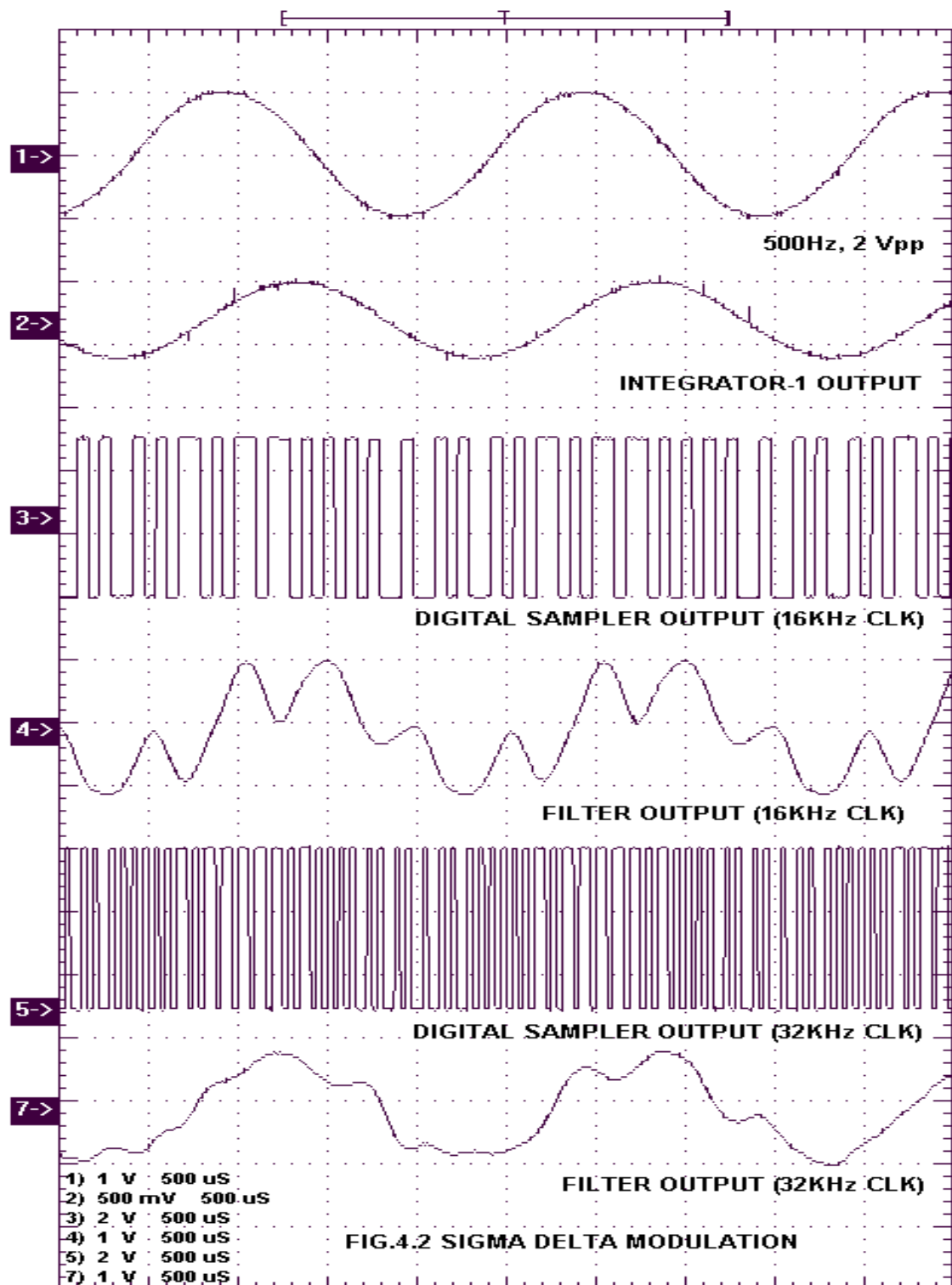
Input Signal

Integrator 1 output

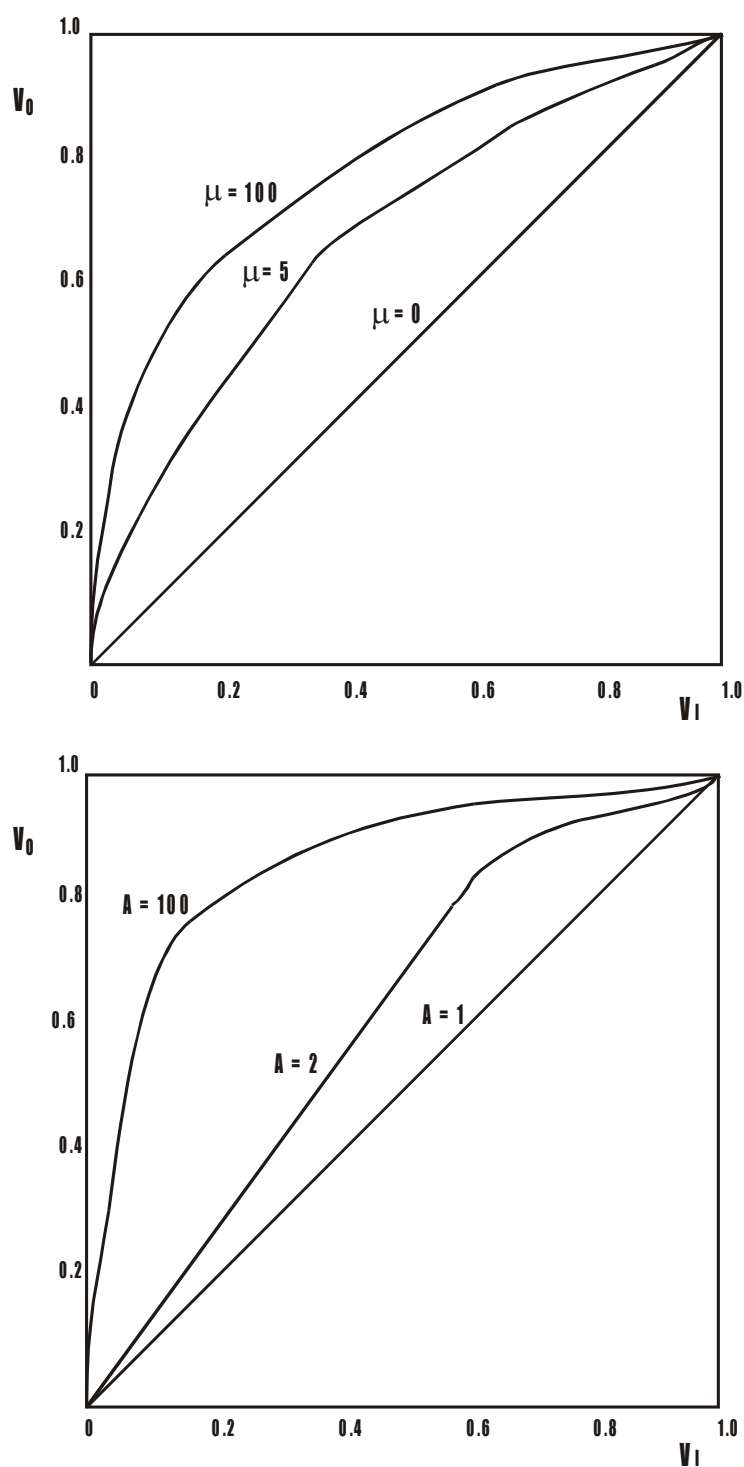
Digital sampler Output

Integrator 2 output

Filter Outputs.



EXPERIMENT NO. 5

**FIG.5.1 COMPRESSION LAWS**

EXPERIMENT NO: 5

NAME:

COMPANDING

OBJECTIVE:

Study of companding system

THEORY:

A process called COMPANDING, a word formed by process of compressing and expanding can materially improve the dynamic range. Companding is the process of compressing the signal amplitudes before transmission and expanding the compressed signal amplitudes after reception.

The dynamic range of Delta Modulation system is limited. The maximum input level is limited by slope overload and minimum input level is limited by quantization noise. This dynamic range can be increased by companding. Companding is accomplished by using Compander, which is consisting of compressor and expander. Companding improves the system performance since the large amplitude input signals can be transmitted and received. The Delta Modulation system encodes the compressed analog signal with fewer errors. The digital signal is decoded and fed as input to the expander. The expander increases the larger amplitudes that had been reduced by the compressor forming the reconstructed analog signal.

In actual form of compression μ -law or A-law is used. In practice μ -law is defined by

$$V_o = \log(1 + \mu V_i) / \log(1 + \mu)$$

And A-law is defined by

$$V_o = AV_i / (1 + \log A) \quad \text{for } 0 \leq V_i \leq 1/A$$

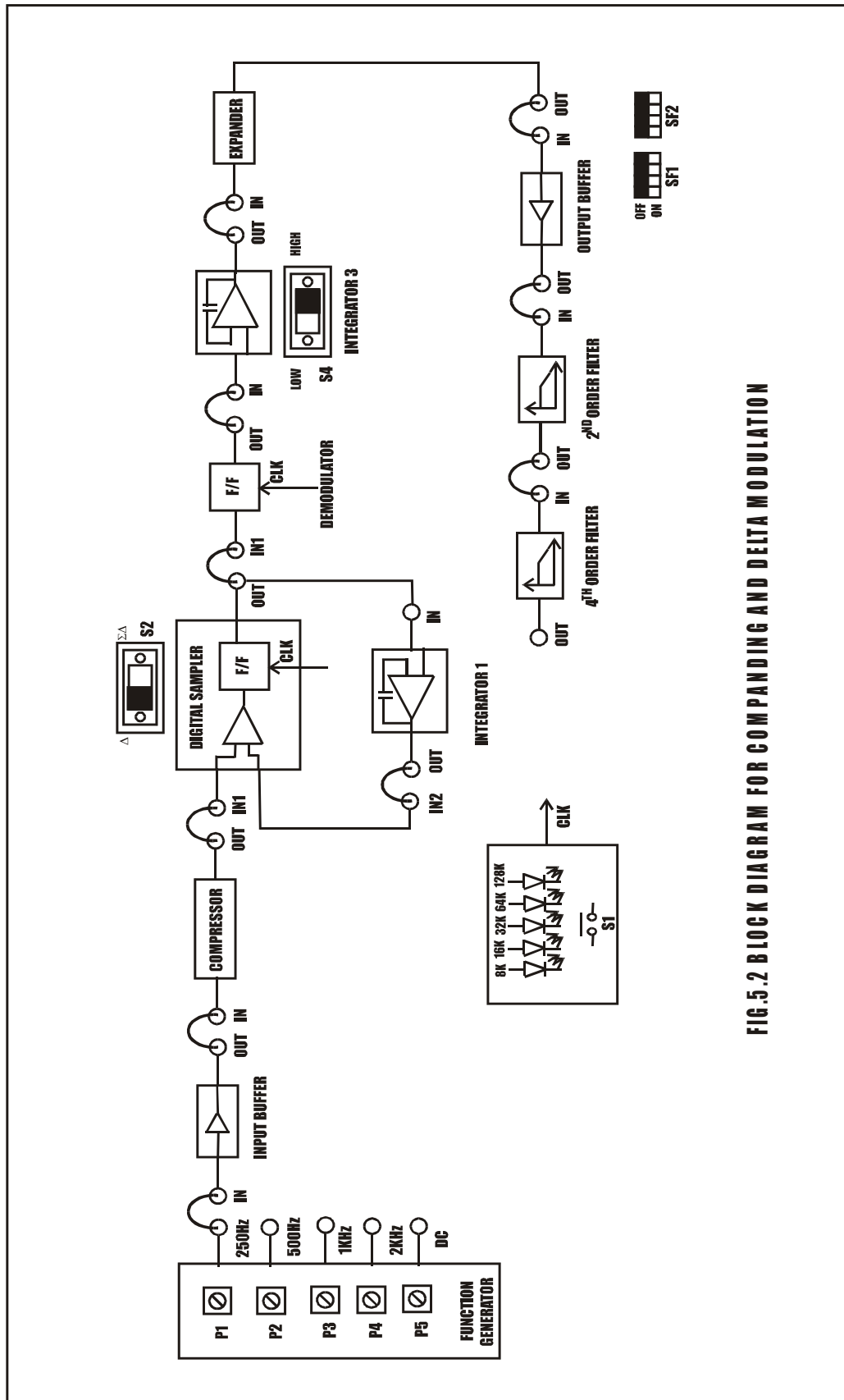
$$= (1 + \log AV_i) / (1 + \log A) \quad \text{for } 1/A \leq V_i \leq 1$$

Where, V_o = Output voltage.

V_i = Input voltage.

μ/A = Positive constant.

Fig.5.1 shows plots for different values of μ/A . The case of uniform quantization corresponds to $\mu = 0$ & $A = 1$. For a given value of μ , the reciprocal slope of the compression curve, which defines the quantum step, is given by the derivative of V_i with respect to V_o . We see therefore that the μ -law is neither strictly linear nor strictly logarithmic, but it is approximately linear at low input levels corresponding to $\mu V_i \ll 1$, and approximately logarithmic at high input levels corresponding to $\mu V_i \gg 1$. Same is the case for A-law.



EQUIPMENTS:

- DCL-07 kit.
- Connecting chords
- Power supply
- Dual Trace Oscilloscope

NOTE: Keep All the Switch Faults (Switch SF1 & SF2) In Off Position

PROCEDURE:

1. Refer to the block diagram (Fig. 5.2) and carry out the following connections.
2. Connect the power supply with the proper polarity to the Kit **DCL-07** and switch it ON.
3. Select sine waves input 250Hz of 2V through pot P1 and connect post **250Hz** to post **IN** of input buffer.
4. Connect output of input buffer post **OUT** to Compressor input post **IN**.
5. Observe the output of compressor as shown in figure 5.2.
6. Connect output of Compressor post **OUT** to Digital Sampler input post **IN1**.
7. Then select clock rate of **16 KHz** by pressing switch **S1** selected clock is indicated by LED glow.
8. Keep Switch S2 in Δ (Delta) position.
9. Connect output of Digital Sampler post **OUT** to input post **IN** of Integrator 1.
10. Connect output of Integrator 1 post **OUT** to input post **IN2** of Digital Sampler.
11. Connect Delta modulated output post **OUT** of Digital Sampler to the input of Delta Demodulator section post **IN** of Demodulator.
12. Connect output of Demodulator post **OUT** to the input of Integrator 3 post **IN**.
13. Connect output of Integrator 3 post **OUT** to the input of Expander post **IN**.
14. Connect output of Expander post **OUT** to the input of output buffer post **IN**.
15. Connect output of output buffer post **OUT** to the input of 2nd order filter post **IN**.
16. Connect output of 2nd order filter post **OUT** to the input of 4th order filter post **IN**.
17. Keep Switch S4 in **HIGH** position.
18. Take Observations at different stages.
19. Then perform the Experiment by changing sine wave frequency to 500Hz and increase the clock frequency to get better reproduction of the input signal.

SWITCH FAULTS:

Note: Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time.

- Put switch **1** of **SF1** in Switch Fault section to **ON** position. This will short the resistor R73 (10K Ω) with R72 (100K Ω), which will reduce Integrator 2 Gain. This will affect the output of Delta / Adaptive Delta Modulation.
- Put switch **2** of **SF1** in Switch Fault section to **ON** position. This will short the resistor R80 (120K Ω) with R81 (1M Ω), which will reduce compression value. This will affect the output of compressor.
- Put switch **3** of **SF1** in Switch Fault section to **ON** position. This will short the resistor R79 (10K Ω) with R78 (100K Ω), which will reduce Integrator 3 Gain. This will affect the output of Delta / Adaptive Delta Demodulation.

- Put switch **4** of **SF1** in Switch Fault section to **ON** position. This will short the generated sine wave signals 250Hz & 500Hz, which will generate noisy sine wave.
- Put switch **5** of **SF2** in Switch Fault section to **ON** position. This will open the bypass capacitor of the 4th order low pass butter-worth filter, which results in the induction of ripples at the filter output.
- Put switch **7** of **SF2** in Switch Fault section to **ON** position. This will open the reset signal of Flip-Flop in Digital Sampler. Results in distorted output at Delta Modulator. Duration of fault gets reduced with higher clock frequency.
- Put switch **8** of **SF2** in Switch Fault section to **ON** position. This will open Resistors R14 & R15 ($22K\Omega$), in ladder network of 250Hz sine wave signal generator. Results in distorted 250Hz sine wave signal.

OBSERVATION:

Observe the following signal on oscilloscope and plot it on the paper. (Fig. 5.3)

Sampling clock

Signal Input.

Compressor Output

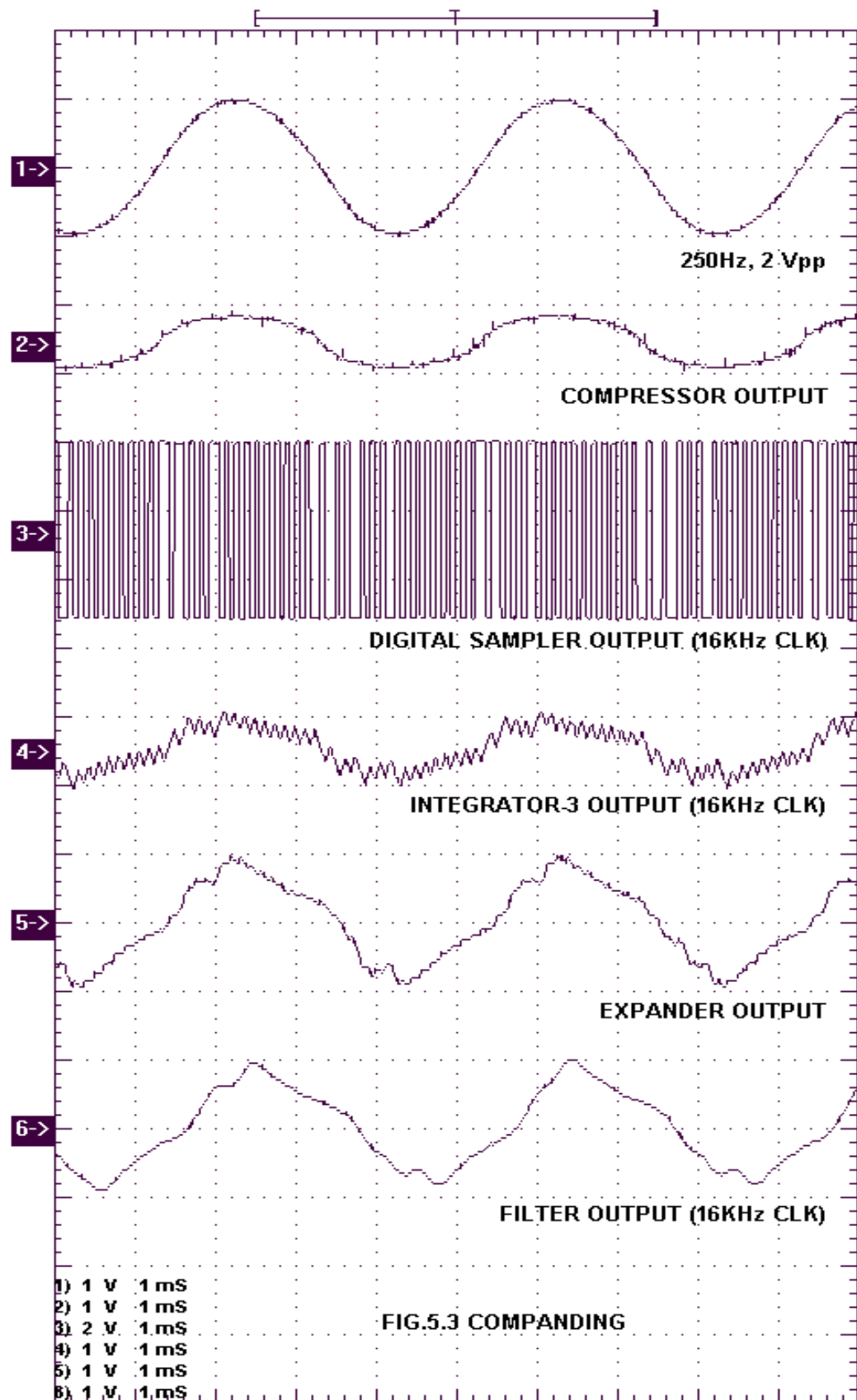
Integrator 1 output

Digital sampler Output

Integrator 2 output

Expander Output

Filter Output.



EXPERIMENT NO. 6

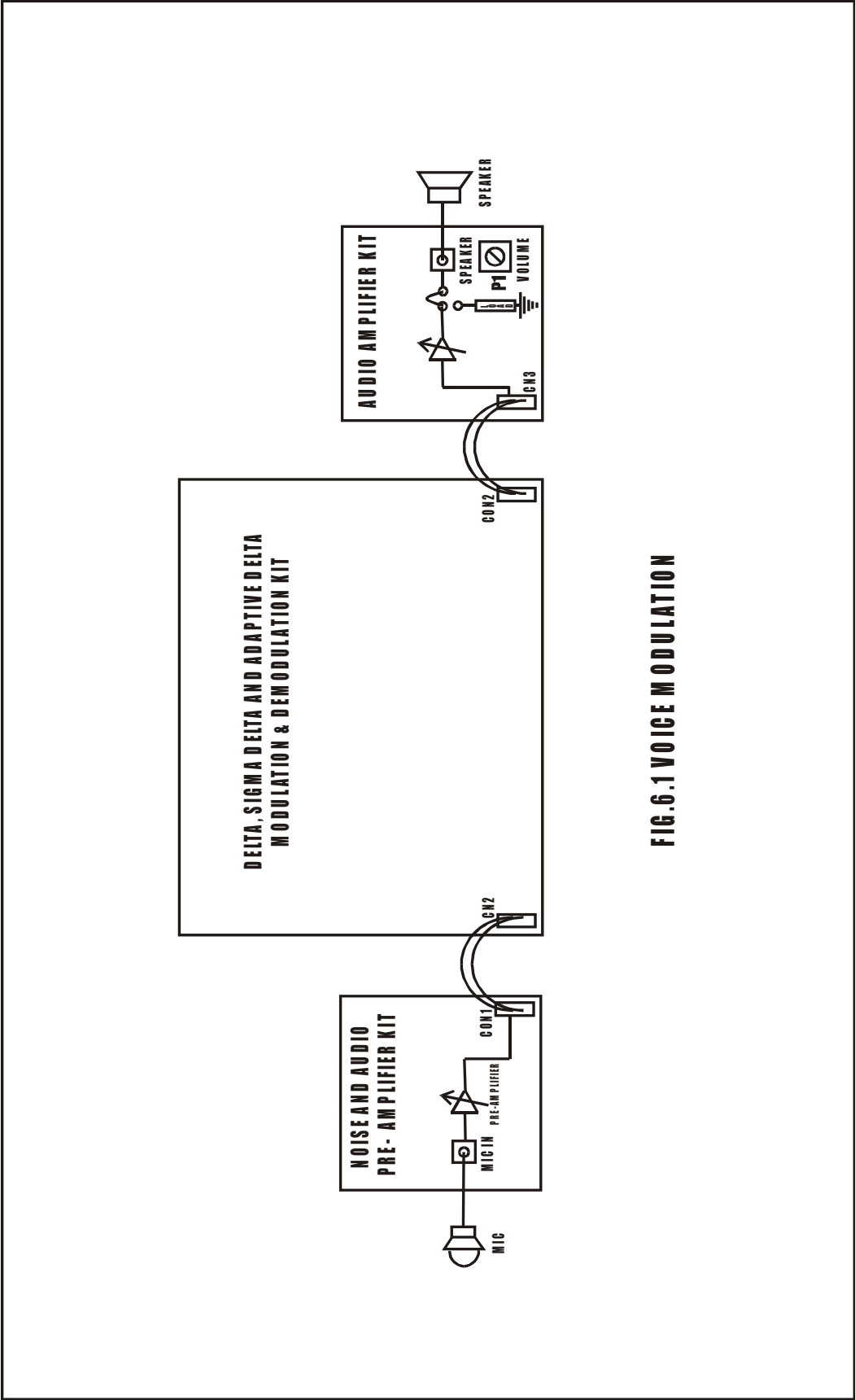


FIG.6.1 VOICE MODULATION

EXPERIMENT NO: 6

NAME:

Voice Modulation

OBJECTIVE:

Study of voice Modulation and Demodulation.

THEORY:

Voice being an Analog signal changes from person to person, One has high pitched voice another a low pitch. One enunciates clearly another slur words to transmit speech we need not transmit the precise waveform generated by speaker. We can transmit information from which a waveform can be reconstructed at the receiver, which is only similar to rather than identical to the waveform generated by the speaker. There is an audio preamplifier before modulation process and audio amplifier after demodulation process. Voice can be modulated by DM, SDM, ADM, CVSD and companding process using Noise and Audio Pre-Amplifier kit & Audio Amplifier kit (Not supplied with DCL-07 Kit).

EQUIPMENTS:

- DCL –07 kit.
- Connecting chords
- Power supply
- Dual Trace Oscilloscope
- Microphone and speaker
- Noise and Audio Pre-Amplifier kit
- Audio Amplifier kit
- Power connector cables

NOTE: Keep All the Switch Faults (Switch SF1 & SF2) In Off Position

PROCEDURE:

PART A: -- Voice Modulation And Demodulation (DELTA)

1. Refer to the block diagram (Fig. 6.1) and carry out the following connections.
2. Connect the power supply with the proper polarity to the Kit **DCL-07** and switch it ON.
3. Connect microphone to the input post **MIC** in Noise and Audio Pre-Amplifier kit.
4. Connect power connection cable from Noise and Audio Pre-Amplifier kit connector **CON1** to DCL-07 kit connector **CN2**.

5. Refer to Fig.1.3 and do the connections of Delta modulation & Demodulation.
6. Remove sine wave signal and connect post **AUDIO IN** to post **IN** of input buffer.
7. Connect post **OUT** of 4th order filter to post **AUDIO OUT**.
8. Connect power connection cable from DCL-07 kit connector **CN3** to Audio Amplifier kit connector **CON2**.
9. Connect speaker at the **SPEAKER** post in audio amplifier Kit.
10. Put Jumper towards speaker post.
11. Observe the quality of reproduction of speech with different clock rates. Adjust the volume knob **P1** to the required level.

PART B: -Voice Modulation And Demodulation (CVSD)

1. Refer to the block diagram (Fig. 6.1) and carry out the following connections.
2. Connect the power supply with the proper polarity to the Kit **DCL-07** and switch it ON.
3. Connect microphone to the input post **MIC** in Noise and Audio Pre-Amplifier kit.
4. Connect power connection cable from Noise and Audio Pre-Amplifier kit connector **CON1** to DCL-07 kit connector **CN2**.
5. Refer to Fig.3.3 and do the connections of CVSD modulation & Demodulation.
6. Remove sine wave signal and connect post **AUDIO IN** to post **IN** of input buffer.
7. Connect post **OUT** of 4th order filter to post **AUDIO OUT**.
8. Connect power connection cable from DCL-07 kit connector **CN3** to Audio Amplifier kit connector **CON2**.
9. Connect speaker at the **SPEAKER** post in audio amplifier Kit.
10. Put Jumper towards speaker post.
11. Observe the quality of reproduction of speech with different clock rates. Adjust the volume knob **P1** to the required level.

PART C: -Voice Modulation And Demodulation (SIGMA DELTA)

1. Refer to the block diagram (Fig. 6.1) and carry out the following connections.
2. Connect the power supply with the proper polarity to the Kit **DCL-07** and switch it ON.
3. Connect microphone to the input post **MIC** in Noise and Audio Pre-Amplifier kit.
4. Connect power connection cable from Noise and Audio Pre-Amplifier kit connector **CON1** to DCL-07 kit connector **CN2**.
5. Refer to Fig.4.1 and do the connections of Sigma Delta modulation & Demodulation.
6. Remove sine wave signal and connect post **AUDIO IN** to post **IN** of input buffer.
7. Connect post **OUT** of 4th order filter to post **AUDIO OUT**.
8. Connect power connection cable from DCL-07 kit connector **CN3** to Audio Amplifier kit connector **CON2**.
9. Connect speaker at the **SPEAKER** post in audio amplifier Kit.
10. Put Jumper towards speaker post.
11. Observe the quality of reproduction of speech with different clock rates. Adjust the volume knob **P1** to the required level.

PART D: --- Voice Modulation And Demodulation (COMPANDING).

1. Refer to the block diagram (Fig. 6.1) and carry out the following connections.
2. Connect the power supply with the proper polarity to the Kit **DCL-07** and switch it ON.

3. Connect microphone to the input post **MIC** in Noise and Audio Pre-Amplifier kit.
4. Connect power connection cable from Noise and Audio Pre-Amplifier kit connector **CON1** to DCL-07 kit connector **CN2**.
5. Refer to Fig.5.1 and do the connections of Delta modulation & Demodulation through companding & expanding.
6. Remove sine wave signal and connect post **AUDIO IN** to post **IN** of input buffer.
7. Connect post **OUT** of 4th order filter to post **AUDIO OUT**.
8. Connect power connection cable from DCL-07 kit connector **CN3** to Audio Amplifier kit connector **CON2**.
9. Connect speaker at the **SPEAKER** post in audio amplifier Kit.
10. Put Jumper towards speaker post.
11. Observe the quality of reproduction of speech with different clock rates. Adjust the volume knob **P1** to the required level.